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PROJECT REPORT ON

VIBRATIONAL CHARACTERISTICS OF FRANCIS TURBINE AT DIFFERENT OPERATIONAL CONDITIONS: A CASE STUDY OF LOWER MARSYANGDI HYDROPOWER PLANT

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PROJECT EVALUATION

Vibrational Characteristics of Francis Turbine at Different Operational Conditions: A Case Study of Marsyangdi Hydropower Plant

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ABSTRACT

This study focuses on the vibration characteristics of a Francis turbine at the Lower Marsyangdi Hydropower Plant under different operating conditions. The turbine was tested at part load (16–18 MW) and full load (21–24 MW). To measure the vibrations, a uniaxial piezoelectric accelerometer sensor with an 8 kHz range and 101 mV/g sensitivity was installed on the turbine guide bearing casing. The electrical signals from the sensor were recorded through an NI-DAQ vibration module, which converted them into raw acceleration data (in g). The entire system was controlled and modeled using LabVIEW, and the collected data was later processed in Python.

The raw signals contained noise and resonance, which were reduced using Butterworth high-pass and low-pass filters. The Nyquist theorem was applied to eliminate repeated cyclic frequencies. After filtering, Fast Fourier Transform (FFT) was used to convert the time-domain data into the frequency domain. This helped identify the vibration frequencies linked to specific turbine issues. To further analyze the signals, acceleration data was integrated to obtain velocity and then displacement using first- and second-order differential equations, which form of concatenated integration. Here, velocity was used to indicate vibration severity, while displacement showed the extent of movement at particular frequencies.

Finally, the frequency signatures were analyzed based on ISO 20816 guidelines to determine safe operating zones and detect potential faults. In addition, the frequency domain plots were studied using measurement data that included draft tube pressure (to observe head changes), RMS velocity (to assess vibration strength), and RMS displacement (to evaluate component movement). The analysis specifically focused on turbine frequency, shaft frequency, and vane passing frequency. This real-time vibration monitoring approach supports condition-based predictive maintenance, helping ensure the reliability and safety of hydropower plant components.

LIST OF ABBREVIATIONS

RBF	Radial Basis Function
SVM	Support Vector Machine
DOF	Degrees of Freedom
FDM	Finite Difference Method
CDM	Central Difference Method
FEM	Finite Element Method
FFT	Fast Fourier Transform
DFT	Discrete Fourier Transform
HHT	Hilbert–Huang Transform
WT	Wavelet Transform
RMS	Root Mean Square
RTD	Resistance Temperature Detector
BEP	Best Efficiency Point
DAQ	Data Acquisition System
VPF	Vane Passing Frequency
CSV	Comma-Separated Values
TDMS	Technical Data Management Streaming
ISO	International Organization for Standardization
mm/s	Millimeters per second
µm	Micrometer
mV/g	Millivolt per g (acceleration)

LIST OF SYMBOLS

M	Mass
C	Damping coefficient
K	Stiffness
X	Displacement
F(t)	External force as a function of time
Hz	Hertz (unit of frequency)
kHz	Kilohertz (unit of frequency)
mm/s	Millimeters per second (velocity)
µm	Micrometer (displacement)
f_s	Shaft frequency
f_t	Turbine frequency
N	Rotational speed (rpm)
N	Number of blades
P	Posterior probability distribution
N	Normal probability distribution
X	Mean
Y	Independent variable
G	Acceleration due to gravity
Rpm	Revolutions per minute

CHAPTER 1 INTRODUCTION

1.1 Background

Hydropower has long been the most important renewable source of electricity, providing clean, reliable, and cost-effective energy [1]. In countries like Nepal, where rivers originating from the Hindu Kush range have been the biggest source of electricity, providing 90% of the total supply [2]. It helps reduce the use of fossil fuels, improves energy security, and supports the country's development. As electricity needs continue to grow and the world moves toward cleaner energy, hydropower plants are expected to work more efficiently and without problems. Similarly, Nepal holds immense potential in producing clean energy through hydropower. Several large-scale projects are currently in the pipeline, aiming to meet the growing energy demands of Southeast Asia. To ensure consistent power production, hydropower plants are equipped with various data monitoring systems. Among these, turbine vibration data is a key indicator of the health of electromechanical components. Monitoring and analyzing this data helps in predicting potential issues and scheduling timely maintenance [3].

The study focuses on the Francis turbine. The source of vibration in this turbine includes irregular water flow, mechanical components, and electrical forces. These vibrations can change depending on how the turbine is running. It is important to understand these vibration patterns to keep the turbine running safely, improve its performance, and avoid serious damage or failure [4]. The source of vibrations in a hydropower plant may be produced by different components. To analyze the faults that arise on the turbines, focusing on mechanical vibration that emerges from several conditions such as mass imbalance, shaft misalignment, bearing irregularities, and structural resonances. The rotating assembly, consisting of the runner, shaft, and generator rotor, is subject to centrifugal force that can amplify any existing imbalance. Manufacturing tolerances, material non-uniformities, and operational wear can all contribute to dynamic imbalance, which typically manifests as vibrations at the rotational frequency [5].

In continuous operation in a fast-flowing river carrying high sediment, the turbine is subjected to erosion. The high sediment content in Himalayan rivers, particularly during

monsoon seasons, creates additional loading on runner blades and can alter the flow patterns, eroding the turbine. Studies conducted on the Middle Marsyangdi Hydropower Plant have documented significant sediment erosion issues, with blade thickness reductions of up to 5 mm observed over four years [6].

Mechanical vibrations in rotating machinery, such as Francis turbines, often produce complex signals containing multiple frequency components, known as vibration harmonics. These signals typically arise from the interaction of dynamic forces—such as hydraulic excitations and mechanical imbalances—with the structural components of the turbine. These vibrations can be detected by accelerometers or displacement sensors mounted on critical components like the turbine guide bearing and generator guide bearing (Kundur, 2021). The raw vibration signal captured over time (in the time domain) can be transformed into the frequency domain using signal processing techniques such as the Fast Fourier Transform (FFT), Discrete Fourier Transform (DFT), Hilbert–Huang Transform (HHT), and Wavelet Transform (WT). Among these, the FFT is the most commonly and widely used method due to its computational efficiency [7]. FFT simplifies the Fourier Transform by breaking the computation into smaller parts and exploiting the symmetry and periodicity of complex exponentials, allowing the conversion of a time-domain signal into its frequency components along with their corresponding amplitudes [8]. This process is run through a graphical technique.

These methods help identify changes in the machine's behavior and determine the health condition of turbine components. For Francis turbines, frequency analysis not only reveals the presence of faults but also helps determine their severity and location, enabling preventive action before failure occurs.

The frequency amplitude spectrum obtained from vibration signal analysis is typically compared against standard reference limits defined by organizations such as ISO 20816. This standard provides guidelines for evaluating the severity of machine vibrations based on measured amplitude levels. If the observed vibration amplitudes exceed or fall significantly below the recommended thresholds, it indicates a deviation from normal operating conditions. In such cases, to prevent further mechanical damage or potential

failure, the turbine unit should be temporarily shut down for thorough inspection and corrective maintenance [9].

Signal filtration is used to clean vibration data by removing extra noise and interference that can hide the real behavior of the system. Raw vibration signals often have unwanted high-frequency electrical noise, random background vibrations, or errors from sampling that are not related to the problem being checked.

A Butterworth filter is often chosen because it keeps the useful signal shape almost unchanged in the range and care about, while removing the unwanted parts. When used together with Fast Fourier Transform (FFT) analysis, the Butterworth filter helps focus on the important frequency range, such as the blade-passing frequency, its harmonics, or slow structural vibrations[10]. This makes it easier to spot problems like cavitation, imbalance, or misalignment. Overall, filtering makes the analysis more accurate by increasing the clarity of the signal, making it easier to detect faults, and reducing the chance of mistakes caused by irrelevant frequency components.

1.2 Objectives

1.2.1 Primary Objective

To determine the vibrational characteristics of the Francis turbine at the Lower Marsyangdi hydropower plant at different operational conditions.

1.2.2 Secondary Objective

1. To study the vibrational characteristics of the Francis turbine.
2. To determine the instrumentation, position, and installation of the vibration sensor.
3. To store and analyze vibration at different operating conditions.

1.3 Scope and Limitation

1. The study is limited to a case study of the vibrational characteristics of the Lower Marsyangdi hydropower plant.
2. The study is limited to the analysis of vibrational data at different operating conditions.

3. A uniaxial accelerometer sensor is used for measuring the vibration data.
4. The study is based on the ISO 20816 standard for time domain analysis and FFT analysis for the frequency domain.
5. Displacement and velocity data are computed from the concatenated integration of acceleration data.
6. The scope of analysis is limited to turbine frequencies of the Lower Marsyangdi hydropower plant.

CHAPTER 2 LITERATURE REVIEW

Major developments in the field of condition monitoring process and its core analysis of vibration to detect faults of turbines are shown in Table 1 below.

Table 1: Literature review on basic information of vibration

S.N.	**Author's Name**	**Major Findings**	**Methods**
01	(Singiresu S. Rao 2010)	• Vibrational behavior • Response characteristics • Vibration Control • Real World application • Multi DOF systems, natural frequency	• Analytical Technique (Newton's law, Lagrange's Equations, Energy methods) • Numerical Methods (FDM, CDM, FEM) • Experimental Technique (use accelerometers, vibrometers, and use of FFT) • Computational Tools (MATLAB, modal analysis)

Table 2: Literature review on the position of sensor installation

S.N	**Reference**	**Instruments used**	**Location of Instruments Installed**	**Significance**	**Sampling frequency (Hz)**
02	(D M Bucur, 2012)	Proximity Transducer	Turbine Shaft	Displacement measurement of a rotating or reciprocating shaft	5000
		Piezoelectric Accelerometer	Draft tube Cone	Measure Vibration on casing or structural absolute motion	
03	(David Valentín, 19 September 2019)	Proximity Probe	Turbine bearing and Generator bearing	Displacement	4096
		Accelerometer	Turbine bearing	Vibration in radial motion	65536
			Generator bearing	Vibration in radial motion	4096
			Thrust bearing	Vibration in axial motion	4096
			Draft tube	Vibration of the draft tube	65536
			Guide vane	Vibration of Guide Vane	65536
			Spiral Casing	Vibration of spiral casing	4096
04	(Luka Selak, 7th Feb 2014) &	Resistance Thermometers (RTDs)	Thrust bearing, upper and lower guide bearing	Measured temperature fluctuation in the thrust	

	(Gunasekara, 2011)			bearing and upper & lower guide bearings	
		Accelerometers	Radial shaft of the turbine	Measured the displacement of the guide vane servo motor and the displacement of the Runner regulating oil distributor.	
		Piezoelectric	Heat Exchanger, Seal Ring and Head Cover, and Hydraulic Systems (Hoists, Valves, Actuators)	Measured pressure of water in the heat exchanger, water pressure in the seal ring, oil pressure of different hydraulic systems, and pressure as the net static head of the Power plant.	
		Electro-vortex sensor	upper guide bearing, axis flange, and turbine guide bearing	Measured radial vibration of the Upper guide bearing, axis flange, and turbine guide bearing.	
05	(British Standard, 2001)	Contact or non-contact displacement transducer	Bearing housing or adjacent surrounding structures	Vibration of non-rotating parts	2-1000

		Seismic transducer or accelerometers	Specific components obtain a flat response to match the sampling rate	Vibration on non-rotating parts	2-1000 or higher

Table 3: Literature review on signal processing of vibration

S.N	**Author's Name**	**Major Findings**	**Methods**
06	Steve Goldman (1991)	• Target frequencies and their significance • Sample frequencies analysis • FFT Spectrum analyzer- How it works? • Transducer for vibration measurement.	• FFT • Numerical methods • Filtration Techniques
07	(Andrew K.S. Jardine, Daming Lin, and Dragan Banjevic 2006)	• Signal Processing (raw condition, Vibration, acoustic, thermal, etc.) • Frequency domain techniques like FFT • Time frequency techniques like the wavelet transform for non-stationary transient signals • Time domain indicators (RMS, kurtosis, skewness) • Hilbert Huang Transform (HHT) Effective signal preprocessing	• Time domain analysis • Frequency domain analysis • Signal Processing Techniques • Data fusing from multiple users

08	Dahal, R., Chitrakar, S., & Neopane, H. P. (2023)	• Frequency Response Shift • Dominant Frequencies • Effect of Load Practical Implication	• Experimental Setup • Vibrational Signal Acquisition • Signal Processing Techniques (FFT, Bandpass filtering, Order analysis)

The reviewed literature highlights various instrumentation techniques and their applications in monitoring the dynamic behavior of hydropower turbine systems.

2.1 Basic of Vibrations

Table 1 shows basic information on vibration as literate below:

- **The book Mechanical Vibrations by S. S. Rao**

S. S. Rao provides a detailed study of vibration analysis, covering both theoretical and practical aspects. It explains the modeling of systems using mass, spring, and damping elements, and uses methods like Newton's law, Lagrange's equations, and energy approaches. The text covers single and multi-degree-of-freedom systems, modal analysis, and numerical techniques such as Runge-Kutta and finite difference methods. MATLAB-based solutions are integrated throughout. It also includes vibration measurement techniques and control methods like damping, isolation, and absorbers [11].

- **Forced vs. Free Vibration**

If the vibrations are caused by an external force or motion, the vibrations are called forced vibrations. Forced vibration happens when something outside the system makes it shake, like water pressure or parts that spin unevenly.

If the vibrations are initiated by an initial energy present in the system and no other source is present, the resulting vibrations are called free vibrations. Free vibration happens when the system shakes on its own, just because it started moving, without anything pushing it [11].

- **Damped vs. Undamped Vibration**

If the vibrations are assumed to have no source of energy dissipation, they are called Undamped. If a dissipation source is present, the vibrations are called damped and are further characterized by the form of damping. Most machines lose energy as they vibrate, which causes the movement to slowly fade [11]. This is called damping. A common type used in studies is viscous damping, which works like a thick liquid, slowing things down. For example, if viscous damping is present, they are called viscously damped.

- **Natural Frequency and Resonance**

Every machine has a natural frequency, a speed it naturally likes to vibrate at. If something pushes the machine at that exact speed, it causes resonance, which makes the shaking much stronger and can lead to damage [11].

These basic ideas help engineers understand and predict how a turbine works in different conditions. They are used to find problems and make the turbine design better by studying its movement and vibrations.

2.2 Selection of location to install sensors

Table 2 shows the literature on the selection of the position of the sensor described below:

- **Bucur (2012)**

Bucur used special sensors called piezoelectric accelerometers on the draft tube to measure vibrations, and proximity sensors on the turbine shaft to track how much it moved. The data was collected 5000 times per second, focusing on how far parts moved and how fast [4].

- **Valentin (2019)**

Valentin used both proximity probes and accelerometers to study vibrations in the spiral casing, draft tube, guide vanes, and bearings. At each location, the sensors could collect data up to 65536 times per second, allowing a very detailed look at vibrations in both straight (axial) and sideways (radial) directions [5].

- **Luka Selak (2014)** and **Gunasekara (2011)**

L. Selak and G. used a variety of sensors to study how turbines behave during operation. They used RTDs (Resistance Temperature Detectors) to measure temperature changes in the bearings, while accelerometers were applied to track the movement of turbine parts. Piezoelectric sensors were used to monitor pressure and fluid systems, and electro-vortex sensors helped detect radial (sideways) vibrations. These investigations mainly focused on understanding the mechanical and thermal behavior of turbine systems [12].

- **The British Standard (2001)**

British Standard provided guidelines for using vibration sensors in turbine systems. It recommended seismic accelerometers and both contact and non-contact displacement sensors to measure vibrations in non-rotating components of the turbine. Depending on the specific application, the standard suggested sampling frequencies ranging from 2 to 1000 times per second, or even higher for more detailed analysis [2].

2.3 Definition of zones and action limits for Francis turbines

- **ISO 20816-5**

Focuses on evaluating machine vibration by measurements on non-rotating parts (bearing housings/pedestals). It classifies machines into 4 groups and provides zone limits (A–D) for vibration severity.

Evaluation Zones:

Zone A: Acceptable for newly commissioned machines

Zone B: Suitable for long-term operation

Zone C: Unsatisfactory for continuous operation

Zone D: Likely to cause machine damage

Table 4: ISO Standard threshold of vibration amplitude

S. N	Zone	RMS Velocity (mm/s)	P-P Displacement (µm)
01	Zone A/B	≤ 1.6	≤ 30
02	Zone B/C	≤ 2.5	≤ 50
03	Zone C/D	≤ 4.0	≤ 80
04	Above D	> 4.0	> 80

The study applies these standards to compare actual measurements with established thresholds [2].

2.4 Vibration Spectrum Analysis

Table 3 shows vibration spectrum analysis and the process of signal interpretation, and is listed below:

- **Steve Goldman (1991)**

S Goldman presented a practical approach to vibration spectrum analysis, emphasizing the identification of target frequencies related to machine components such as shafts, bearings, and gears. Goldman highlighted the importance of selecting appropriate sampling frequencies to prevent aliasing and explained the functioning of FFT spectrum analyzers for converting time-domain signals into frequency-domain data. Goldman also discussed the use of transducers like accelerometers, velocity sensors, and displacement probes, stressing their selection based on application needs, and also laid the foundation for effective vibration-based condition monitoring and fault diagnosis in rotating machinery [10].

- **Andrew K.S. Jardine, Daming Lin, and Dragan Banjevic (2006)**

Andrew K.S Daming Lin and D. Banjevic present a detailed framework on condition-based maintenance with a strong focus on signal processing techniques used in machinery diagnostics and prognostics. Their study explores the application of time domain indicators such as RMS, kurtosis, and skewness, along with frequency domain methods like the Fast Fourier Transform (FFT) and advanced time-frequency techniques, including wavelet transforms, particularly suited for analyzing non-stationary and transient signals. They also discuss the Hilbert-Huang Transform (HHT) for extracting meaningful features from nonlinear and non-stationary data. The work emphasizes the importance of effective signal preprocessing and data fusion from multiple sources to improve diagnostic accuracy. Time and frequency domain analyses are both applied to interpret signals related to vibration, acoustic emissions, and thermal conditions [13].

- **Dahal, Chitrakar, and Neopane (2023)**

Dahal, Chitrakar, and Neopane investigate the impact of varying load conditions on the vibrational behavior of hydropower turbines. Their study identifies shifts in frequency response and highlights dominant frequencies under different operational states. The methodology includes an experimental setup for acquiring vibration signals, followed by signal processing using FFT, bandpass filtering, and order analysis. This research offers practical insights into how load affects turbine dynamics, supporting improved condition monitoring and fault detection [14].

2.5 Signal Processing

In signal processing, different techniques are used, like FFT, HHT, WT, and STFT. These techniques use different solving methods. Different signal processing techniques are used to transform time-domain signals into the frequency or time–frequency domain, each using a distinct method. The Hilbert–Huang Transform (HHT) applies Empirical Mode Decomposition followed by the Hilbert transform to analyze nonlinear and nonstationary signals adaptively. The Wavelet Transform (WT) decomposes signals into scaled and shifted wavelets, providing a multi-resolution view suitable for nonstationary and transient events. The Short-Time Fourier Transform (STFT) segments the signal into short windows and applies FFT to each, yielding a spectrogram but with a fixed trade-off between time and frequency resolution [15].The Fast Fourier Transform (FFT) uses an efficient algorithm to compute the Discrete Fourier Transform, making it fast and reliable for stationary signals [16].

- **FFT**

It is one of the common methods used for the conversion of a time domain signal into its frequency domain spectrum. According to **R.B. Randall in Frequency Analysis (3rd edition, 1987)**, the Fast Fourier Transform (FFT) is preferred among other signal transformation techniques due to its significant advantages in both efficiency and accuracy. FFT drastically reduces the computational complexity of the Discrete Fourier Transform (DFT) from $O(N^2)$ to $O(N \log N)$, making it feasible to analyze large datasets quickly and even in real-time applications. It provides high-resolution spectral analysis, allowing precise identification of frequency components, which is especially useful for detecting subtle features in vibration and acoustic signals. Additionally, the widespread adoption of FFT has made advanced digital analyzers more affordable, increasing accessibility for a variety of industrial and research applications. Its proven reliability, accuracy, and integration into standard measurement systems have established FFT as a foundational and widely accepted method in frequency analysis, ensuring consistency and comparability of results across different platforms.

- **Signal aliasing**

Signal aliasing occurs when a signal is sampled too slowly, meaning the data points are collected at a rate that is not fast enough to capture the true variations of the signal. According to the **Nyquist-Shannon sampling theorem**, to accurately represent a signal, it must be sampled at least twice as fast as its highest frequency. If this rule is not followed, higher-frequency components can appear as lower-frequency ones, distorting the signal and leading to incorrect analysis. Aliasing can cause false frequency readings, which in applications like mechanical vibration analysis may result in wrong conclusions about the condition of machinery. To prevent aliasing, engineers either increase the sampling rate to exceed twice the highest frequency or apply an anti-aliasing filter, a low-pass filter that removes high-frequency components before sampling [17]. These measures ensure that the sampled signal accurately reflects the original, avoiding misinterpretation and potential errors in signal analysis.

- **Broadband Frequency**

Broadband frequency vibration refers to vibration signals that span a wide range of frequencies, rather than being concentrated at a few discrete frequencies. Francis turbines operate under varying hydraulic loads and off-design conditions, which can introduce broadband vibration patterns. The broadband frequencies are introduced from sources like hydraulic excitation, runner faults, and sediment erosion. These frequencies can be cut off or removed using the Nyquist-Shannon sampling theorem, cutting the cyclic frequency of the sampling rate and Fast Fourier transform by cutting off the low-pass and high-pass frequencies. And need the ten times of the base frequency of 10 to 1000KHz to read the radial frequency vibration, including misalignment, imbalance, and turbine frequencies.

CHAPTER 3 METHODOLOGY

3.1 Study Design

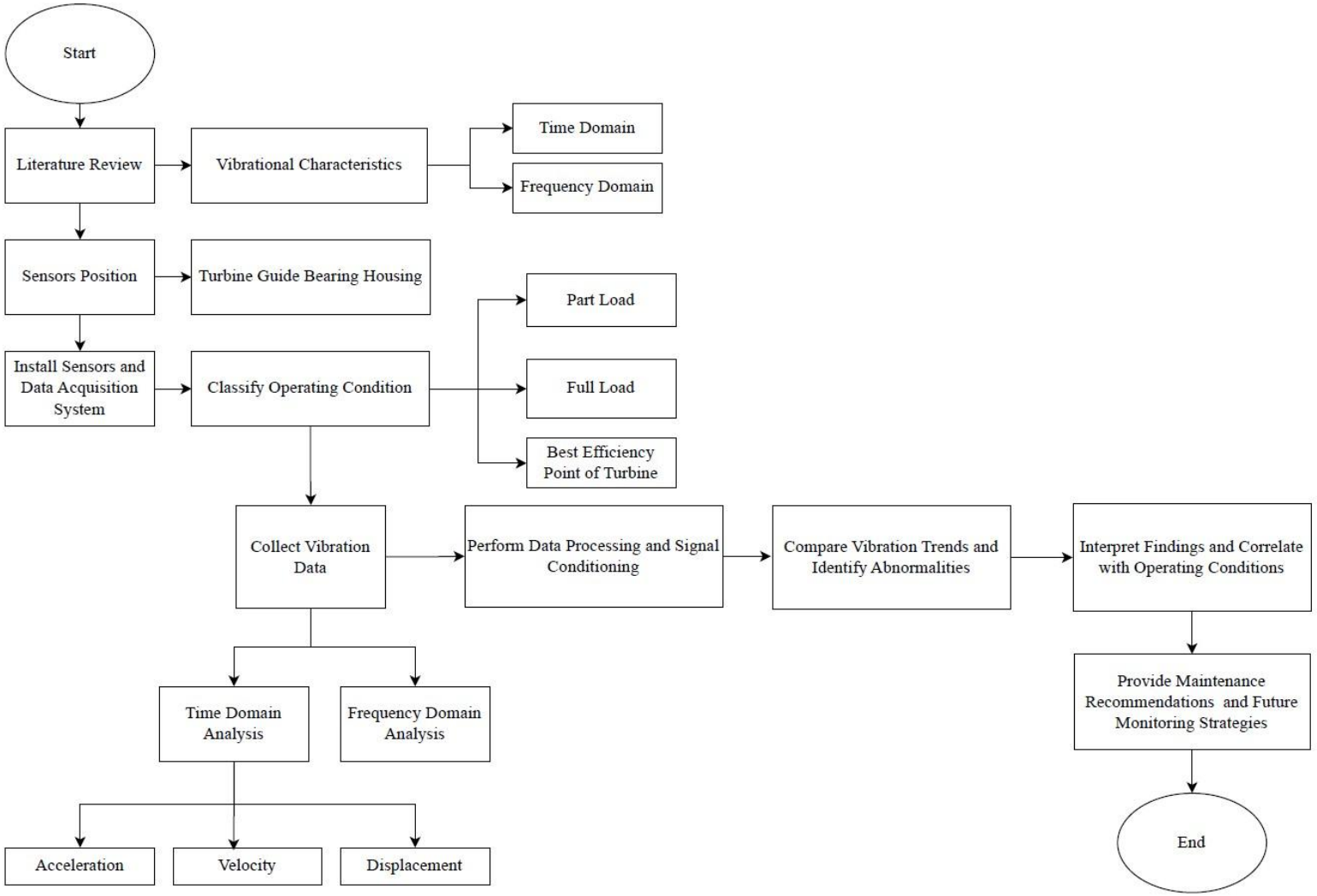


Figure 1: Detailed Methodology Flowchart

Figure 1: A Detailed methodology flow chart of the project is described to carry out the case study in vibrational characterization of the Francis turbine. At first, the literature review on various practices of condition monitoring by vibrational frequency and amplitude in hydropower stations is conducted. Now, a specific location for the sensor placement is identified according to the sensor's usability and literature. The appropriate location for data observation in the mechanical parts of the Lower Marsyangdi Hydropower is the Upper Guide Bearing and the Turbine Guide Bearing. The sensor was tuned with the data acquisition system and vibrational module, which performs condition-based monitoring and collects the data from the site, and sends it to the server for further analysis. The vibrational data for the same mechanical component's characteristics was closely observed in all load conditions: part, full, and base loads. The data processing is done by FFT (Fast Fourier Transform) in Python, after the graphical data is interpreted carefully on the basis of the standard trend.

After the study, it was possible to assess the health of the turbine and other hydromechanical Components, and maintenance is recommended according to the fault detection.

3.1.1 Sensor Position

Based on the literature review, piezoelectric accelerometers are commonly installed at various locations within hydropower plants to measure critical vibrational parameters. These include the turbine guide bearing, generator guide bearing, and thrust bearing to monitor radial and axial motions. Additionally, accelerometers are placed on the radial shaft of the turbine to measure the displacement of the guide vanes and the runner.

3.2 Tools and Techniques

3.2.1 Sensor

Various types of sensors can be used for vibration monitoring in hydropower plants, including displacement sensors, velocimeters, and accelerometers. These sensors are available in different specifications and functions, and the selection often depends on the available space and the specific measurement requirements.

At the Lower Marsyangdi Hydropower Plant, due to the limited space available at the turbine guide bearing casing, a uniaxial piezoelectric accelerometer was selected as the most

suitable option. It is mounted by drilling a hole of a size of diameter equal to the sensor diameter with an interference of a tight fit. The chosen accelerometer has a measurement frequency range up to 8 kHz and a sensitivity of 101 mV/g, making it appropriate for detecting high-frequency vibrations associated with turbine operation.

Figure 2: Installed piezoelectric accelerometer sensor [19]

3.2.2 Sound and Vibration Module

The Sound and Vibration Module detects audio signals and physical vibrations. It uses microphones to capture sound and sensors to pick up movement or impact, enabling devices to respond to noise, touch, or environmental changes. The NI C series 4-channel vibration module is installed on the site as shown in Figure 5.

Figure 3: Installed vibration module[19]

3.2.3 DAQ

A DAQ (Data Acquisition System) is essential for analyzing the vibrational characteristics of a Francis turbine by collecting and converting analog signals from accelerometers into digital data for analysis. When accelerometers are mounted on the turbine guide bearing, they detect vibrations in various directions, and the vibration module conditions these signals by amplifying and filtering them.

Figure 4: Data Acquisition System[19]

The DAQ system digitizes the conditioned signals and sends them to analysis software for further processing. One of the most commonly used tools for this purpose is LabVIEW. This integration of DAQ hardware with LabVIEW software provides a flexible and powerful platform for conducting efficient and accurate vibration analysis, ultimately supporting predictive maintenance and ensuring the reliable performance of the turbine.

3.2.4 LabView

LabVIEW is a graphical programming environment that captures, analyzes, and displays data in real time. It interfaces with sensors and data acquisition hardware to measure equipment conditions. With built-in signal processing capabilities and customizable front panels, LabVIEW allows for rapidly developing scalable and efficient monitoring systems. It works by making a graphical algorithm to control and monitor the sensor and module for signal processing. In terms of this project, it helps to illustrate the graphical interface of the vibration monitoring program, and is used to process the displacement signal and interact with the DAQ and vibration module to develop fine data.

3.2.5 Python

The process of analyzing vibration signals using Python for applications such as condition monitoring of Francis turbines involves several systematic steps. Initially, vibration data is collected from accelerometers mounted on critical components of the machinery. This data is typically acquired through a data acquisition (DAQ) system and may be logged directly using Python-compatible DAQ libraries or indirectly via software like LabVIEW. When LabVIEW is used, the recorded data can be exported in formats such as CSV or TDMS and then imported into Python for processing.

Once the acceleration data is available in Python, it undergoes preprocessing to remove unwanted noise and isolate the relevant frequency bands. This is typically achieved using a Butterworth bandpass filter, which allows frequencies within a specified range to pass while attenuating frequencies outside that range. The filtered signal is then analyzed in both the time and frequency domains. In the frequency domain, the Fast Fourier Transform (FFT) is applied to convert the time-based signal into its spectral components, revealing the dominant frequencies present in the vibration. These frequencies are compared against known mechanical fault frequencies such as the ball pass frequency of the inner and outer raceways of bearings, vane passing frequency, and shaft rotational frequency. Identifying peaks at these characteristic frequencies can indicate specific faults such as bearing damage, unbalance, or hydraulic instabilities.

Simultaneously, time-domain analysis is conducted by integrating the filtered acceleration signal to derive velocity and then displacement. Since integration can introduce baseline drift due to low-frequency noise, a high-pass filter is often applied to the integrated signals to maintain accuracy. Once acceleration, velocity, and displacement signals are obtained, peak-to-peak values are calculated for each to quantify the severity of vibration. These measurements help in assessing the operational health of components and diagnosing early signs of wear or failure.

Finally, the processed signals are visualized using plots for acceleration, velocity, and displacement in the time domain, as well as frequency spectra to highlight dominant vibration modes. This comprehensive approach allows engineers to perform detailed diagnostics of rotating equipment, such as Francis turbines, and supports proactive

maintenance strategies by detecting abnormal vibration patterns linked to specific mechanical faults.

3.3 Conceptual Framework

Vibration is the repetitive back-and-forth motion of an object around its equilibrium point. It is produced due to imbalances, misalignments, looseness, and worn-out components. Transducers and displacement sensors measure it. Vibration produces harmonics, which are multiples of the frequency generated by the vibration. The amplitude of this frequency indicates the strength of its signal and helps identify problems such as misalignment and wear of rotating parts [2].

Mechanical vibration theory fundamentally studies the oscillatory motion of mechanical systems about an equilibrium point, describing the underlying principles of how and why structures or machines vibrate. At its core, vibration theory formulates the motion via mathematical models, often starting with a simple spring-mass-damper system, which is governed by the differential equation:

$$m\frac{d^2x}{dt^2} + c\frac{dx}{dt} + kx = F(t)$$

Where m is the mass, c is the damping coefficient, k is the stiffness, x is the displacement, and F(t) is an external force varying with time.

Vibrations are classified by the number of degrees of freedom necessary for their modeling, the type of forcing they are subject to, and the assumptions used in the modeling. Vibrations of systems that have a finite number of degrees of freedom are called discrete systems. A system with one degree of freedom is referred to as a single-degree-of-freedom (SDOF) system. A system with two or more degrees of freedom is called a multiple degree-of-freedom (MDOF) system. A system with an infinite number of degrees of freedom is called a continuous system or a distributed parameter system

Understanding the fundamentals of mechanical vibration is essential for analyzing the dynamic behavior of hydraulic turbines. Mechanical vibrations are periodic oscillations of mechanical components and are typically characterized based on:

3.4 Vibration in Hydraulic Turbines

Francis turbines are widely used in hydropower plants, like the one at Lower Marsyangdi. These turbines work with water, mechanical parts, and electrical systems, so they experience vibration due to multiple factors:

- **Hydraulic Forces:**

When the turbine runs at part load or outside its ideal range, phenomena such as turbulence, pressure changes, swirling water in the draft tube, and unstable flow can cause strong vibrations.

- **Mechanical Causes:**

Problems like an unbalanced shaft, contact between rotating and fixed parts, or misalignment can also lead to vibration.

- **Electromechanical Effects**

If the generator has a magnetic imbalance, it can create vibrations in the turbine too.

Operational conditions (partial load, full load, and best load) greatly affect the vibrational behavior. Understanding how these conditions influence vibration helps in setting alarm/trip limits.

3.5 Condition Monitoring and Vibration Analysis Methods

Vibration-based condition monitoring is a predictive maintenance tool used to assess machine health and detect faults. The following methods are used:

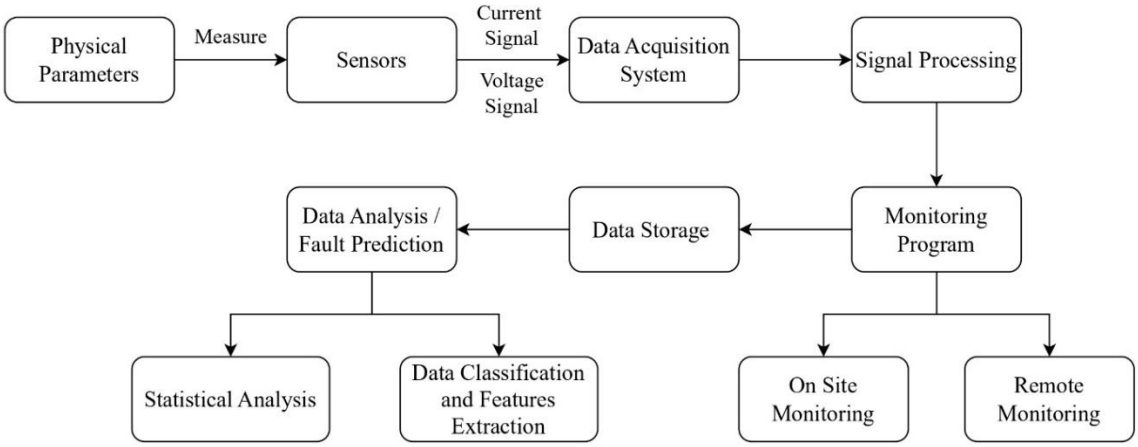


Figure 5: Condition Monitoring Process

- **Time-Domain Analysis:**

Time domain analysis means studying how a signal changes over time, with time on the x-axis and the signal (like vibration, velocity, or displacement) on the y-axis. It is useful for seeing sudden changes, trends, or patterns and for calculating values like RMS, peak, and peak-to-peak. This allows real-time observation of vibration behavior.

- **Frequency-Domain Analysis (FFT):**

Frequency domain analysis uses FFT (Fast Fourier Transform) to convert a time-based signal into its frequency components. In this form, the x-axis shows frequency, and the y-axis shows the amplitude of each frequency. It helps identify the exact sources of vibration, such as shaft speed, vane passing, or turbine frequency. This approach enables decomposition of the signal into individual frequency components.

- **Broadband vs. Narrowband Monitoring:**

Broadband RMS values are used for severity assessment, while narrowband helps in fault diagnosis. These methods help analyze the measured signals and understand how they change under different operating conditions.

- **LabView**

LabVIEW for Vibration Monitoring is a graphical programming environment that captures, analyzes, and displays vibration data in real time. It interfaces with sensors and data acquisition hardware to measure equipment conditions. With built-in signal processing capabilities and customizable front panels, LabVIEW allows for rapidly developing scalable and efficient vibration monitoring systems

In a Francis turbine at a hydropower plant, different hydromechanical components such as guide vane, guide vane casing, turbine shaft, guide vane bearing, misalignment of the shaft, and erosion can induce vibration [18]. This fault may cause failures, increase overall maintenance costs, and decrease the efficiency of the power plant. Vibration analysis, integrated with condition-based maintenance, serves as an essential method for addressing this issue, which introduces predictive maintenance and is implemented at the hydropower plant to enhance operational reliability. Sensors like proximity displacement sensor, piezoelectric accelerometer, and seismic transducer are mounted on rigid parts of the bearing housing or adjacent surrounding structures, which can be classified to give a representative vibration response of the turbine [12]. A time domain signal generated from the sensor is collected by using a vibration module and a DAQ system [19]. This condition monitoring system can be monitored in real-time using a time-based data system, using a server data storage system through LabVIEW software [20].

- **Signal Conditioning:**

Includes amplification and filtering to ensure accurate signals

- **Sampling Frequency:**

It should be at least twice the highest frequency of interest (Nyquist criterion), typically between 2 Hz to 1 kHz for turbine applications.

The placement of sensors at bearing pedestals and critical points ensures comprehensive monitoring of turbine behavior.

3.5 Operational Behavior of Francis Turbines

Francis turbines operate efficiently over a range of heads and flows but show varied behavior across operational zones:

- **Best Efficiency Point (BEP):**

The Turbine shows minimal vibration and optimal hydraulic performance. It is a condition where a power plant operates at design discharge.

- **Partial Load:**

Flow separation, vortex shedding, and pulsations may increase vibration. It is a load that is generated if the available discharge is lower than the design discharge. The part load has been taken as 16-18 MW

- **Full Load:**

Transient conditions induce random excitations due to flow unsteadiness.

Knowledge of these zones is vital for interpreting vibration trends and ensuring machine integrity. It is a load that is generated if the available discharge is greater than the design discharge, and the full load is taken as 21 – 24 MW

CHAPTER 4 RESULTS AND DISCUSSION

4.1 Vibrational Characteristics of Francis Turbine

Vibrational Characteristics of a Francis Turbine (Lower Marshyangdi HPP)

Understanding the vibrational characteristics of Francis turbines is important because these machines operate under heavy hydraulic loads, and any abnormal vibration can signal a developing fault. From the literature, vibration can be studied both theoretically and practically. In theory, vibration analysis begins with simple models of a turbine represented as masses, springs, and dampers. Using methods like Newton's laws of motion, Lagrange's equations, and energy approaches, researchers can describe how the turbine responds to disturbances. These models help identify natural frequencies, damping behavior, and resonance conditions, which are the backbone of understanding how the machine behaves in operation. Rao (2010) emphasizes that single-degree and multi-degree-of-freedom systems provide a foundation for predicting vibration response, and numerical techniques like finite difference and finite element methods refine these predictions for complex structures like turbines.

Another theoretical aspect is the classification of vibrations into free and forced types. Free vibration occurs when the turbine is disturbed once and then allowed to vibrate on its own, while forced vibration happens when an external force, such as water pressure or imbalance, keeps driving the vibration. Damping, which is the gradual loss of energy as the system vibrates, is another key element because it influences how long vibrations last and how severe they become. Resonance, when external forces match the natural frequency of the turbine system, can amplify vibrations to dangerous levels. These theoretical principles provide engineers with guidelines for predicting risky conditions before they appear in practice.

Literature also highlights how theoretical knowledge is applied to real-world turbine monitoring. Researchers like Steve Goldman (1991) show that vibrations in turbines are linked to identifiable frequencies, such as shaft rotation speed, blade-passing frequency, or draft tube vortex rope oscillations. By knowing what frequencies to expect, engineers can set up monitoring systems that compare measured vibrations with theoretical predictions. Signal processing methods like the Fast Fourier Transform (FFT) translate raw vibration

data into frequency spectra, making it easier to match real signals with theoretical models. Advanced techniques like wavelet analysis or Hilbert–Huang Transform help when signals are non-stationary, such as during load changes.

In summary, the theoretical basis for studying Francis turbine vibrations comes from mathematical modeling, classifications of vibration types, and the concepts of damping, resonance, and natural frequency. Literature shows that when these theories are combined with modern signal processing, they provide powerful tools for understanding and predicting how turbines behave under different operating conditions. For a plant like Lower Marshyangdi HPP, applying this knowledge means engineers can design and plan better monitoring systems and extend the lifespan of critical components.

4.2 Instrumentation, Position, and Installation of Vibration Sensor

In this study, vibration measurements at the Lower Marsyangdi Hydropower Plant were carried out using two uniaxial piezoelectric accelerometers, each with a frequency range of up to 8 kHz and a sensitivity of 101 mV/g. The sensors were mounted at 90° to each other on the turbine guide bearing casing to record radial vibrations along both the X and Y axis. This position was selected because the guide bearing is directly influenced by shaft motion

Figure 6: Position of accelerometer sensor installed

and hydraulic forces, making it a critical location for capturing representative vibration signals. Due to site-specific constraints, this location was selected for optimal accessibility. The accelerometers were firmly installed by drilling and tight-fitting to ensure good contact with the casing surface and minimize signal loss due to poor coupling. The recorded analog signals were transmitted to a four-channel NI vibration module connected to a data acquisition (DAQ) system, where they were digitized and processed through LabVIEW before being exported for further analysis in Python.

4.3 Signal Processing

A vibration sensor (uniaxial accelerometer) was attached to the turbine guide bearing casing to measure vibration levels. The sensor produced signals as current and voltage, which were collected using a data acquisition system. Figure 7 shows the processed frequency spectrum generated using Python. The spectrum was plotted from the raw vibration data recorded on June 27, with a sampling frequency of 12,800 Hz and a recording duration of 12 seconds. After being processed, these signals were converted into acceleration data and sent to the server at Kathmandu University. Using Python, the time-based vibration signals were then converted into frequency information with the help of the Fast Fourier Transform (FFT). Finally, the results were shown as graphs of frequency versus amplitude for both the X and Y directions, as illustrated in Figure 7. The frequency spectrum of the complex signals generated by machines is characteristic of each machine, representing a unique pattern, referred to as the "machine signature" [21].

4.3.1 Description of Plot

The horizontal axis in both subplots represents frequency in hertz (Hz), spanning from 0 to approximately 600 Hz, while the vertical axis shows the normalized amplitude of the FFT output. The upper subplot corresponds to the X-axis measurement and is plotted in purple, while the lower subplot corresponds to the Y-axis measurement and is plotted in blue. Both spectra include three annotated reference frequencies, highlighted with vertical dashed lines: the shaft rotational frequency at approximately 5 Hz (orange), the turbine frequency at about 65 Hz (green), and the vane passing frequency (VPF) at approximately 120 Hz (red). These markers serve as critical diagnostic indicators, allowing visual correlation between known mechanical events and peaks in the measured spectrum.

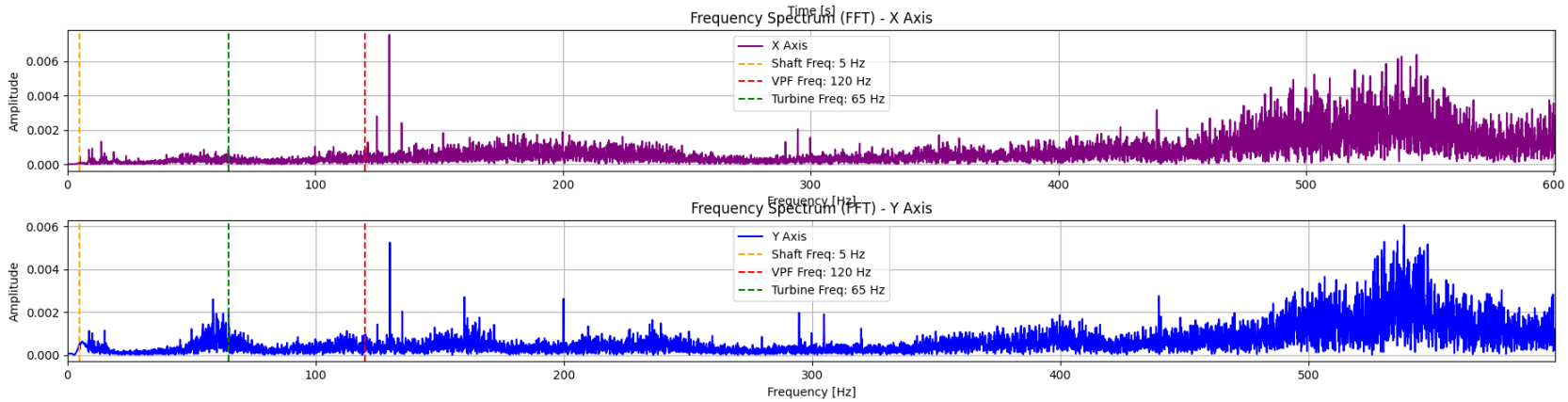


Figure 7: Processed frequency spectrum using python

In the X-axis spectrum, the amplitude is relatively low across the lower frequency range, with distinct but modest peaks at the shaft frequency and its harmonics. The turbine frequency at 65 Hz appears as a small but noticeable peak, suggesting that turbine-related periodic forces are transmitted into this measurement direction. A more prominent response occurs near the VPF frequency at 120 Hz, indicating a strong interaction from stationary vane structures or similar periodic aerodynamic impacts. Above 500 Hz, there is a significant rise in broadband spectral content, which may correspond to turbulent aerodynamic excitation, structural resonance, or the presence of high-speed contact or friction phenomena.

The Y-axis spectrum exhibits a somewhat different distribution of energy. Low-frequency content, particularly around the shaft frequency, is more pronounced than in the X-axis measurement, indicating a stronger sensitivity in this direction to rotational imbalance or shaft motion. The peaks at 65 Hz and 120 Hz are again present, but the energy is more dispersed, suggesting a broader frequency spread of excitation in this axis. The most notable difference is the markedly stronger broadband energy beyond 500 Hz, which may be associated with asymmetric aerodynamic loading, lateral vibration modes, or mounting and structural coupling effects that preferentially excite this axis.

4.3.2 Analysis of vibration signature

From a diagnostic standpoint, the narrow peaks at the reference frequencies are characteristic of periodic excitation linked to known mechanical components, while the broadband high-frequency content likely arises from stochastic or turbulence-driven sources. The asymmetry between the X-axis and Y-axis responses highlights the importance of multi-axis measurement, as it reveals directional dependencies in vibration behavior that can be critical for fault identification. While overall amplitudes remain relatively low, the spectral distribution provides valuable insights into the mechanical and aerodynamic interactions within the system, allowing targeted investigation of components operating near the identified frequencies.

- **Low-frequency shaft line (5 Hz)**

The shaft order at 5 Hz appears as a distinct but low-amplitude line. A low, well-defined shaft line generally indicates that the rotor imbalance or synchronous shaft motion is present but modest. The lack of large harmonics or strong modulation sidebands around 5 Hz suggests no severe dynamic unbalance or gross shaft misalignment at the moment of measurement.

- **Turbine frequency (65 Hz)**

A visible peak at 65 Hz indicates normal turbine forcing (blade/turbine natural excitation or blade pass harmonic). Because the peak amplitude is moderate and not accompanied by large sidebands or a forest of nearby discrete lines, there is no immediate evidence of a resonant excitation strongly amplifying the 65 Hz content. This is consistent with expected operation but should be trended.

- **Vane Passing Frequency**

The VPF at 120 Hz is a noticeable feature in both X and Y spectra and appears stronger than the shaft and primary turbine lines. The presence of 120 Hz energy and any visible harmonics/multiples is characteristic of periodic forcing by stator vanes or wicket gates acting on the flow or blades. A relatively high VPF amplitude indicates that vane–blade interactions or flow pulsations are a dominant excitation source for the machine. If the peak is stationary, it is likely a hydraulic excitation (expected in Francis turbines). If it shows amplitude modulation or sidebands, that would point to interaction with a rotating structural mode (potential rubbing, looseness, or modulation by a slower process).

Overall, the turbine does not show signatures of an acute, catastrophic mechanical fault in these snapshots. The shaft line is small (limited imbalance), and the turbine line is moderate. The dominant diagnostic finding is a relatively strong VPF at 120 Hz and pronounced high-frequency broadband energy (especially in Y). This combination points to hydraulic excitation and possible early hydraulic/structural interactions rather than immediate rotor catastrophic failure. However, the elevated VPF and broadband high-frequency content are indicators that warrant attention because they can drive fatigue in blades/vanes, accelerate wear on seals and bearings, or indicate cavitation activity that erodes components over time

4.3.3 Sampling rate and cutting frequency

The sampling period is 0.000078125 seconds, which means a “cutting frequency” of 6400 Hz (Nyquist frequency). Figure 7 represents the FFT plots of the vibration signal acquired with the accelerometer sensor. In order to identify the vibration sources, the peaks of the measured frequency spectra are correlated with the data related to the possible component sources in the machine. This requires knowledge of the operating and geometrical parameters, such as rotational speed, number of blades, etc. [22]. There are also the frequency of the shaft, Vane Passing Frequency, and Turbine Frequency.

The frequency corresponding to the synchronous rotational speed of the analyzed turbine, which has a value of 300 rpm.

$$f_s = \frac{300}{60} = 5Hz$$

Where, f_s = shaft frequency

The fundamental frequency or the turbine frequency is around 65Hz, which corresponds to the synchronous rotational speed multiplied by the number of turbine blades (13 blades)

$$f_t = \frac{n}{60} \times N = \frac{300}{60} \times 13 = 65\ Hz$$

Where, f_t = turbine frequency

And the Vane Passing Frequency of the turbine is around 120 Hz.

$$VPF = \frac{Number\ of\ vanes \times RPM}{60} = \frac{24}{60} \times 300 = 120\ Hz$$

Figure 7 presents a comparative analysis of vibration signals in the frequency domain using Fast Fourier Transform (FFT) techniques for two orthogonal measurement directions, the X-axis and the Y-axis. The objective is to identify dominant vibration components, correlate them with known mechanical excitation sources, and assess the presence of broadband or harmonic features that may indicate specific fault mechanisms.

4.3.4 Comparative analysis of displacement over the ISO guideline

Table 5: Comparison with ISO 20816-5 Vibration Severity Limits

Parameter	**Measured Value**	**ISO Limit (Zone A/B)**	**Compliance**	**Remarks**
X-axis RMS Velocity	0.458531 mm/s	≤ 1.6 mm/s	Within limit	Very low vibration level, well inside Zone A/B range.
X-axis P–P Displacement	12.1517 µm	≤ 30 µm	Within Limit	Significantly below limit, no concern.
Y-axis RMS Velocity	0.511217 mm/s	≤ 1.6 mm/s	Within Limit	Low vibration, stable condition.
Y-axis P–P Displacement	12.1517 µm	≤ 30 µm	Within Limit	Well within safe range.

Table 5 shows a comparison of the produced displacement with standard data

According to ISO 20816-5, the measured RMS velocities (0.46 mm/s in X-axis and 0.51 mm/s in Y-axis) and peak-to-peak displacements (12.15 µm for both axis) are well below the Zone A/B thresholds. This places the turbine's current operating condition in the Good to Acceptable for long-term operation range, indicating minimal mechanical stress and no signs of significant imbalance, misalignment, or excessive hydraulic excitation.

The low vibration levels suggest that the post-overhaul condition of the turbine is excellent, and the machine is running smoothly under the current load and hydraulic conditions. Continuous monitoring is still recommended, especially during high-sediment monsoon periods, to detect early changes that might trend toward Zone B/C levels.

4.4 Shaft displacement analysis

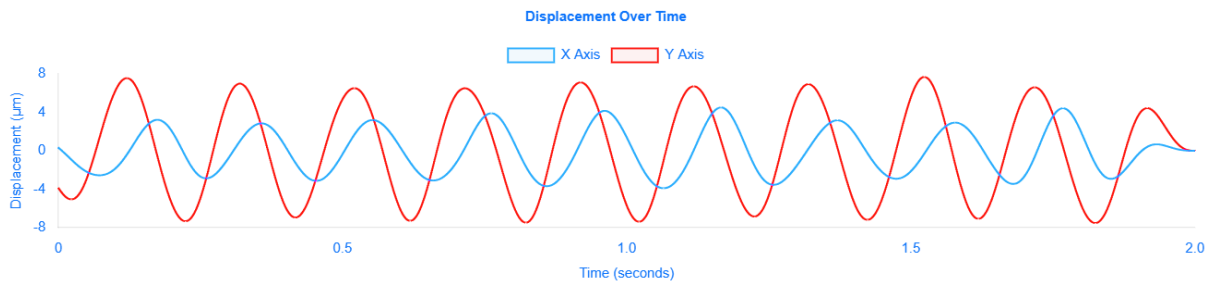


Figure 8: P-P Displacement of Shaft

Figure 8 shows the peak-to-peak displacement of the shaft frequency, whose amplitude is shown on the Y axis and its time is shown on the X axis, and all the displacements are below 30 µm, which is below the ISO guideline, and the plant shaft is in a safe condition.

4.5 Store and analyze the vibration data at different operating conditions

The vibration data collected from the Lower Marsyangdi Hydropower Plant are directly stored in the Kathmandu University server, which can be accessed anytime from within the university premises. The data are saved in two formats: frequency data and processed measurement data. The measurement data include timestamp, RMS velocity of the X and Y axis, peak-to-peak displacement of the X and Y axis, acceleration data of both axis, as well as frequency and amplitude information for the X and Y axis.

4.5.1 Comparative study of vibration frequency spectrum at different operating conditions

From the 1st week of June, the monsoon season in Nepal begins, during which hydropower plants operate at their highest potential because of the increased river flow. In this project, while analyzing the data and log sheets of the Lower Marsyangdi Hydropower Plant, it was observed that the plant mainly operated at maximum loads of 22 MW and 20 MW during the project observation period. According to the recorded data, the highest operation levels during this season were achieved in the months listed in Table 6.

Table 6: Different operating conditions in the specified time

S.N.	Month	Operating condition
1	June	18MW
2	July	22MW
3	August	22MW

- **4.5.1.1 Comparative study of the spectrum of 25th June and 4th July on 21MW Power**

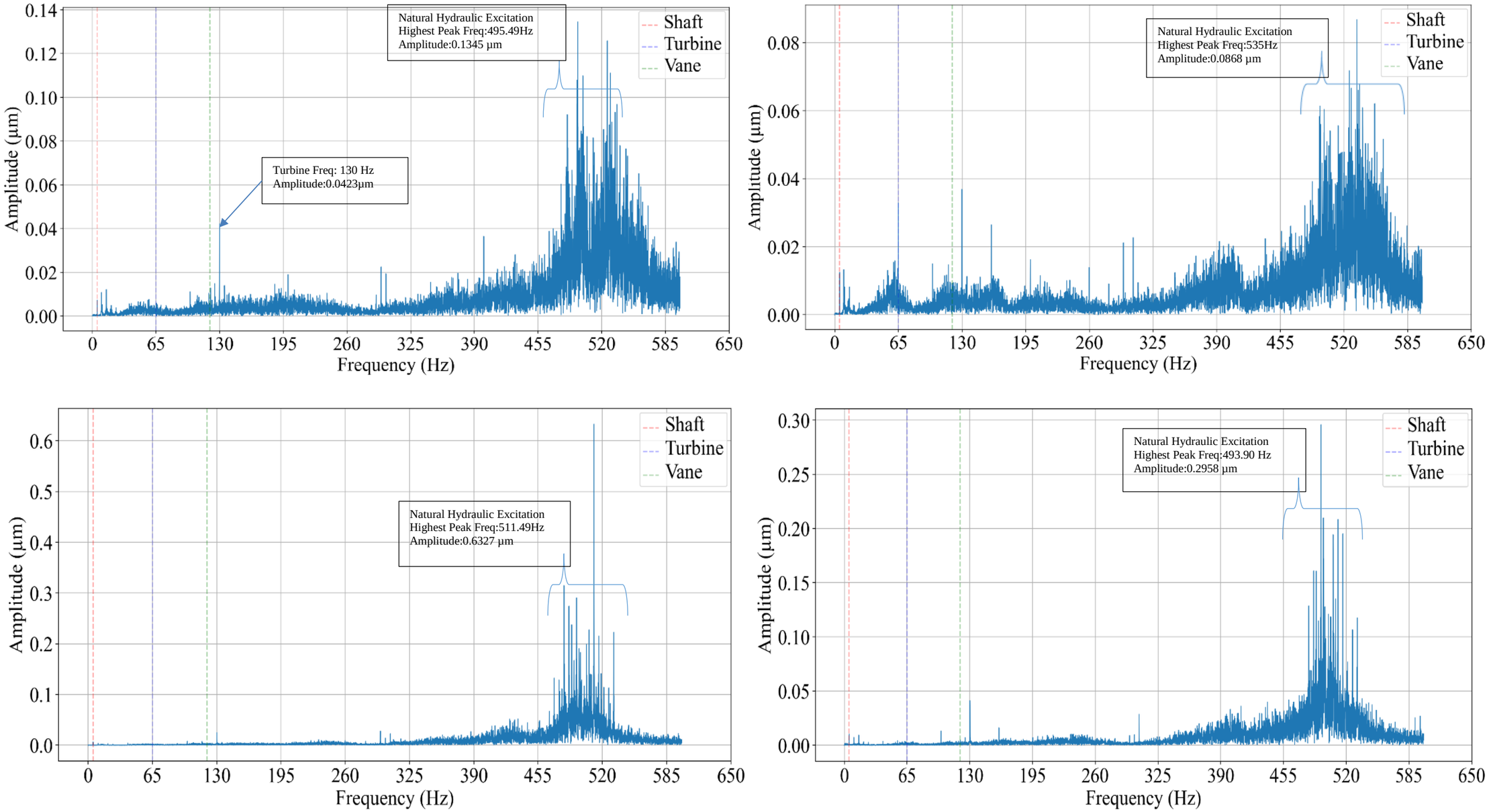


Figure 9: Comparative study of spectrum of 25th June and 4th July of 21 MW

- **X-Axis Spectrum of 25th June**

On 25th June, the vibrations in the X-axis were comparatively slightly greater than those of the Y-axis vibrations. The main peak was around 435Hz, with an amplitude of only about 0.1345 µm. This level indicates that what is mostly normal water-flow effects, not strong resonance or mechanical problems. This is a low value, which means the turbine was running smoothly without any serious vibration issues. The other frequencies linked to turbine speed and vane passing did not show much change either. Overall, the X-axis results on this day suggest the machine was in a stable and healthy condition. As the natural Hydraulic Excitation is greater in amplitude and earlier than the Y-axis, this shows the vibration in the X-axis is slightly greater than in the Y-axis.

- **Y-Axis Vibration, 25th June (21 MW)**

The Y-axis on 25th June tells a similar story. The vibration peak was also near 535 Hz, and the amplitude was less than 0.0868 µm. Just like the X-axis, the Y-axis vibrations were low and steady, showing that the unit was running safely.

- **Peak-to-peak displacement**

The peak-peak displacement is lower in the Y-axis as compared to the X-axis. Whereas, the draft tube pressure seems to be uniform at about 1.5 in average. The RMS Accelerations, in the Y-axis, are smaller where peak-to-peak displacement is about double in value as compared to the x-axis, i.e., 16 µm in average.

- **X-Axis Vibration 4th July (21 MW)**

The pattern seems different on 4th July. In the X-axis, a strong vibration peak appeared at around 511 Hz, reaching an amplitude of 0.6327 µm. This is almost five times higher than observed on 25th June. The sharp, high-frequency vibration suggests that the turbine entered into a condition called hydraulic resonance, where water flow patterns and pressure inside the draft tube vary, and cause vibration in the structure. This is not dangerous on its own, but it shows the machine was being pushed into a much more energetic vibration state than the day before.

- **Y-Axis Vibration, 4th July (21 MW)**

On the Y-axis, the same trend was seen, though at a lower level than the X-axis. The main peak was at 493.90 Hz, with an amplitude of 0.2958 µm. Even though this is smaller than the X-axis peak, it is still several times higher than the values recorded on 25th June. This means that both

directions of vibration were affected by the same flow-induced resonance, with the X-axis showing the stronger spectrum.

From the data of July 4, the essential frequency other than the natural excitation frequency seems to be similar. This signifies, the problem is not in the vanes, turbine, or shaft alignment but due to a certain change in the draft tube pressure as compared to the previous operating condition. Also, RMS Acceleration is increasing in the X-axis compared to the Y-axis, which is shown in Figure 10 below. The peak-to-peak displacement is higher in the Y-axis. Due to some issue, the plant was shifted to 22MW after running for 30 minutes

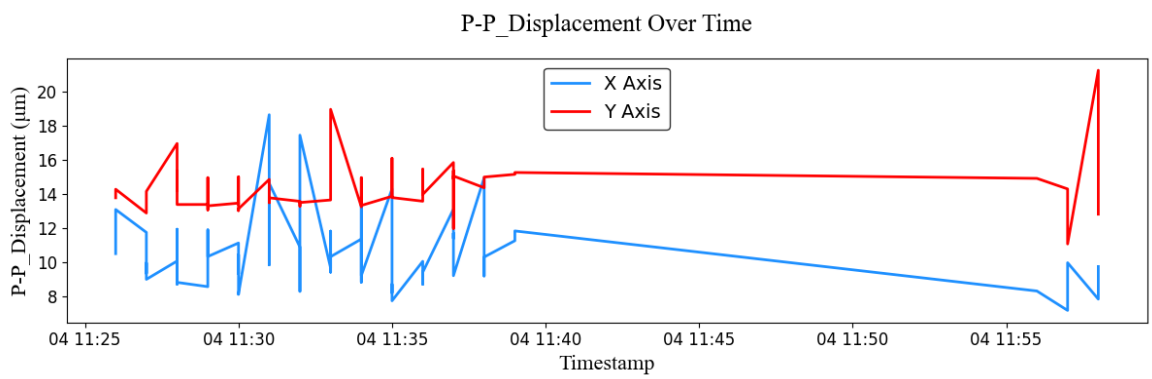


Figure 10: Graphical Representation of RMS Acceleration Vs Timestamp 4th July

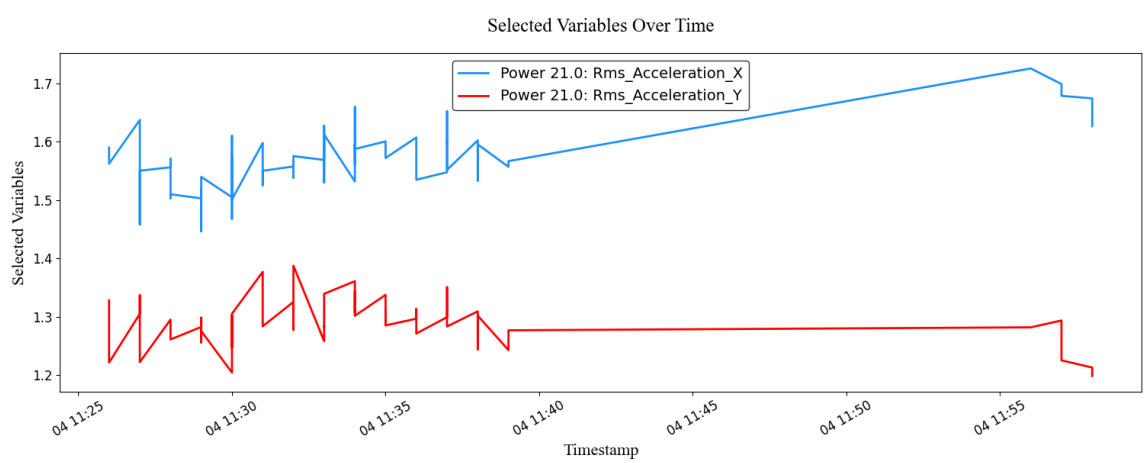
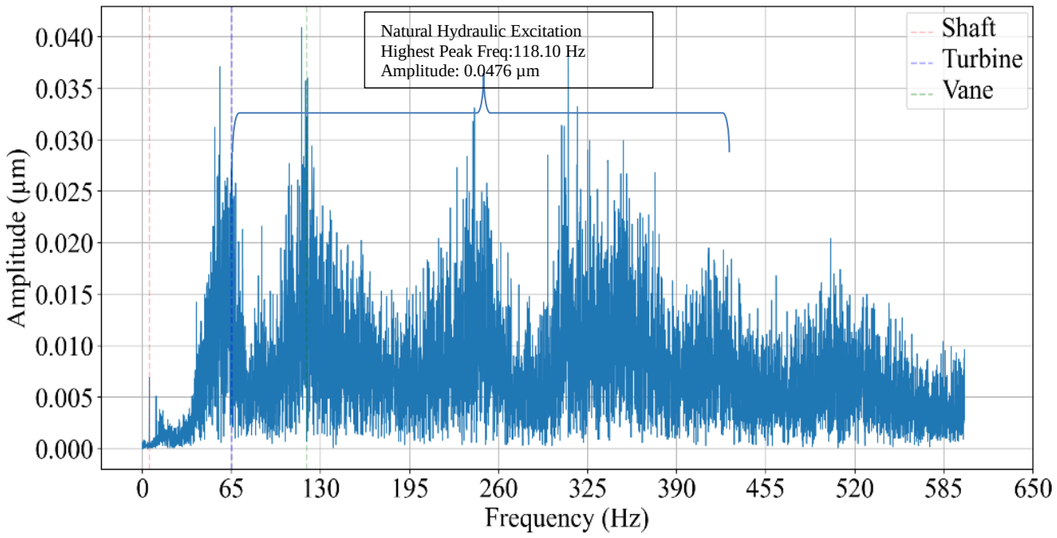
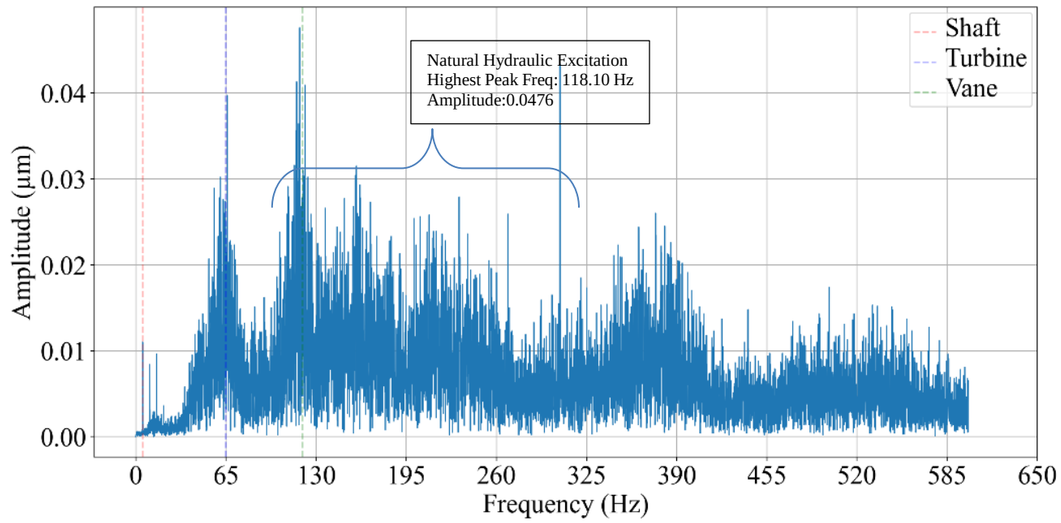
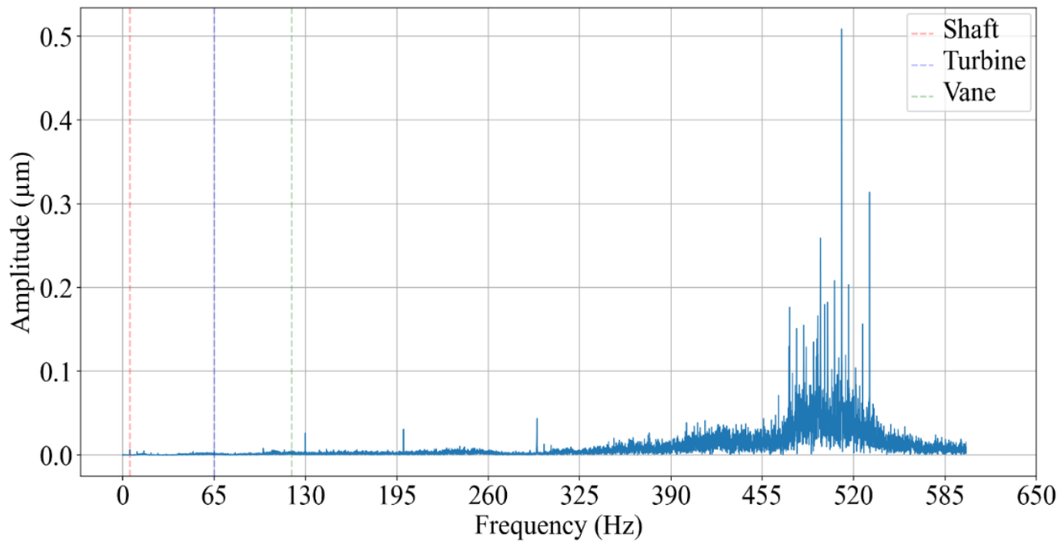


Figure 11: Graphical Representation of P-P Displacement Vs Timestamp 4th July

- **4.5.1.2 Comparative study of the spectrum of 25th June and 4th July on 21MW Power**







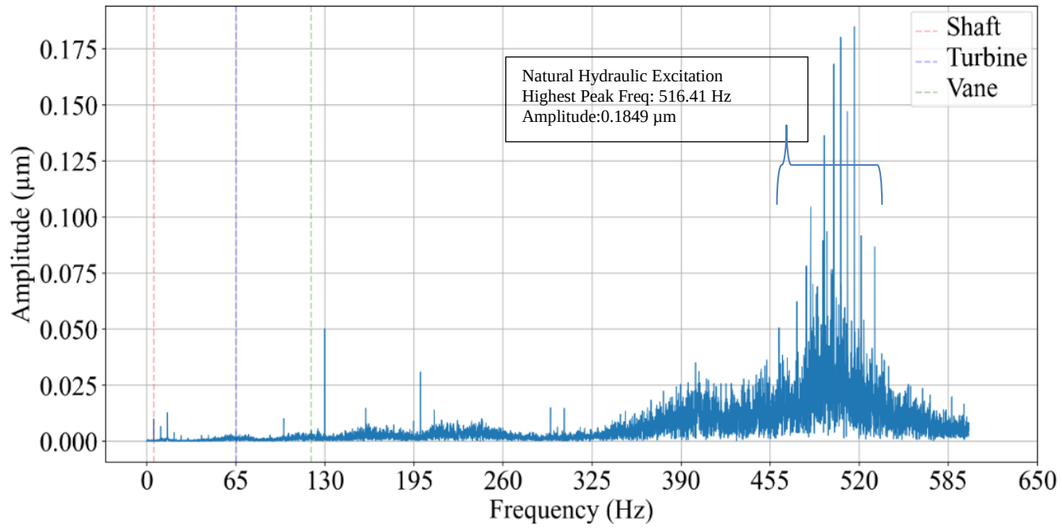


Figure 12: Comparative study of the spectrum of 26th June and 27th June of 22 MW

- **X-Axis Vibration, 26th June (22 MW)**

On 26th June, the vibrations in the X-axis were very small. The main peak was around 118 Hz, with an amplitude of only about 0.047 µm. This is a very low value, which means the turbine was running smoothly without any serious vibration issues. The other frequencies linked to turbine speed and vane passing did not show much change either. Overall, the X-axis results on this day suggest the machine was in a stable and healthy condition.

- **Y-Axis Vibration, 26th June (22 MW)**

The Y-axis on 26th June tells a similar story. The vibration peak was also near 118 Hz, and the amplitude was less than 0.05 µm. This low level indicates that what is seen is mostly normal water-flow effects, not strong resonance or mechanical problems. Just like the X-axis, the Y-axis vibrations were low and steady, showing that the unit was running safely. The peak-peak displacement is higher in the Y-axis as compared to the X-axis. Whereas, the draft tube pressure seems to be uniform at about 1.5 in average.

- **X-Axis Vibration 27th June (22 MW)**

The pattern seems different on 27th June. In the X-axis, a strong vibration peak appeared at around 511 Hz, reaching an amplitude of 0.509 µm. This is almost ten times higher than observed on 26th June. The sharp, high-frequency vibration suggests that the turbine entered into a condition called hydraulic resonance, where water flow patterns inside the draft tube excite the structure. This is not dangerous on its own, but it shows the machine was being pushed into a much more energetic vibration state than the day before.

- **Y-Axis Vibration, 27th June (22 MW)**

On the Y-axis, the same trend was seen, though at a lower level than the X-axis. The main peak was at 516 Hz, with an amplitude of 0.185 µm. Even though this is smaller than the X-axis peak, it is still several times higher than the values recorded on 26th June. This means that both directions of vibration were affected by the same flow-induced resonance, with the X-axis showing the stronger response.

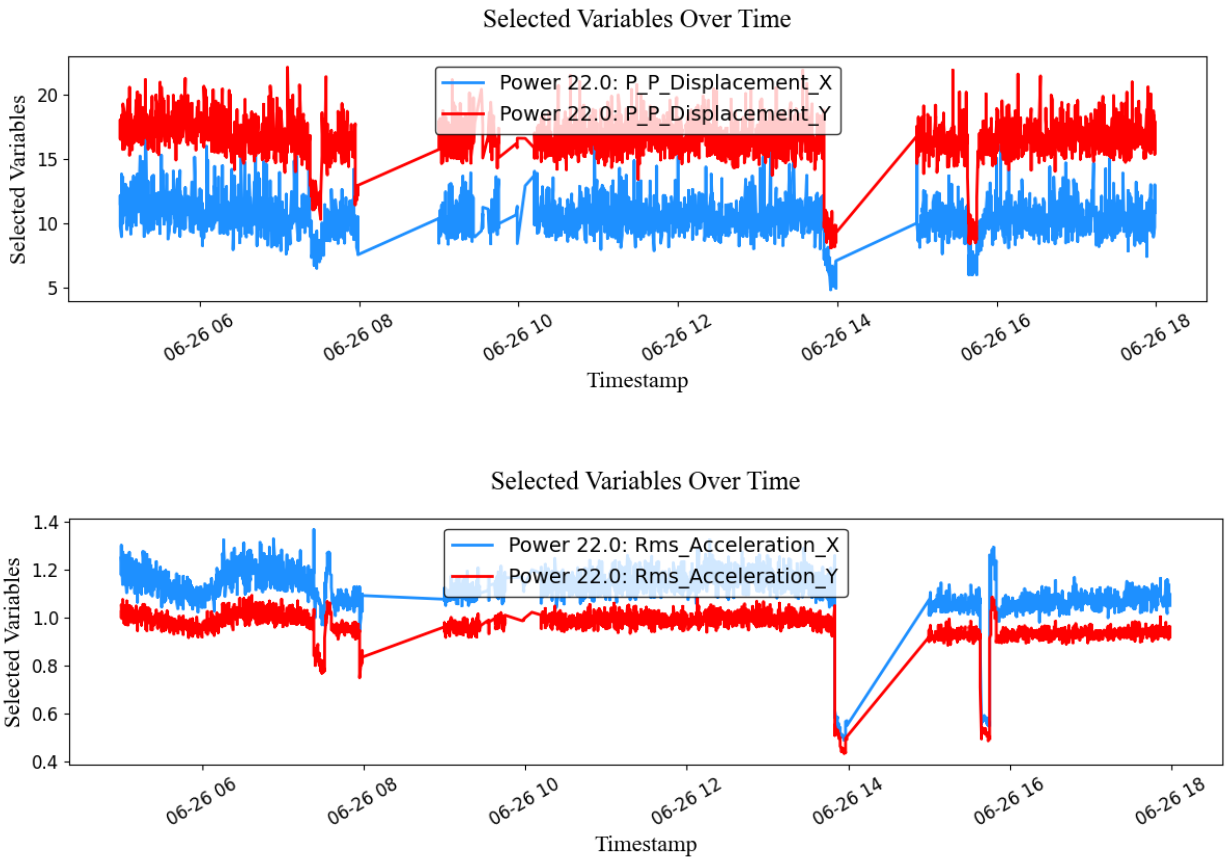


Figure 13: P-P displacement of 26th June

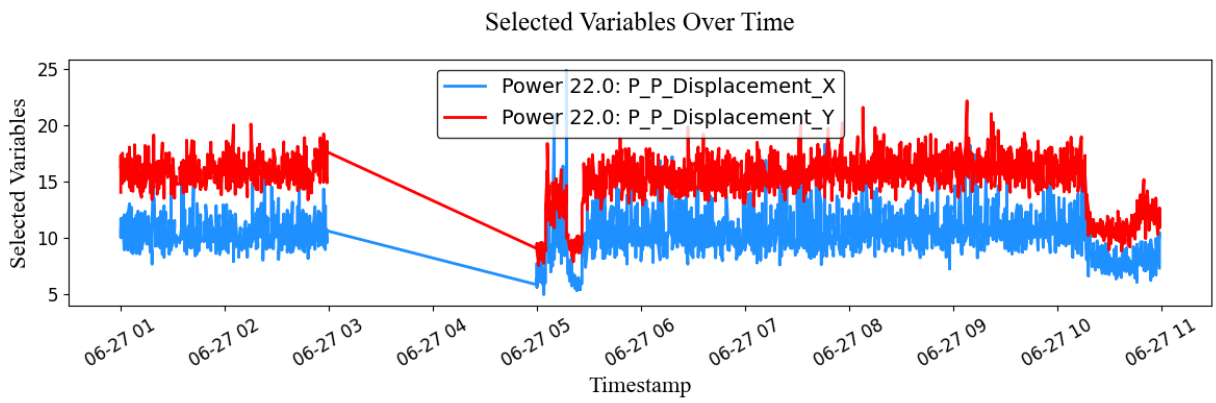


Figure 14: P-P displacement 26th June

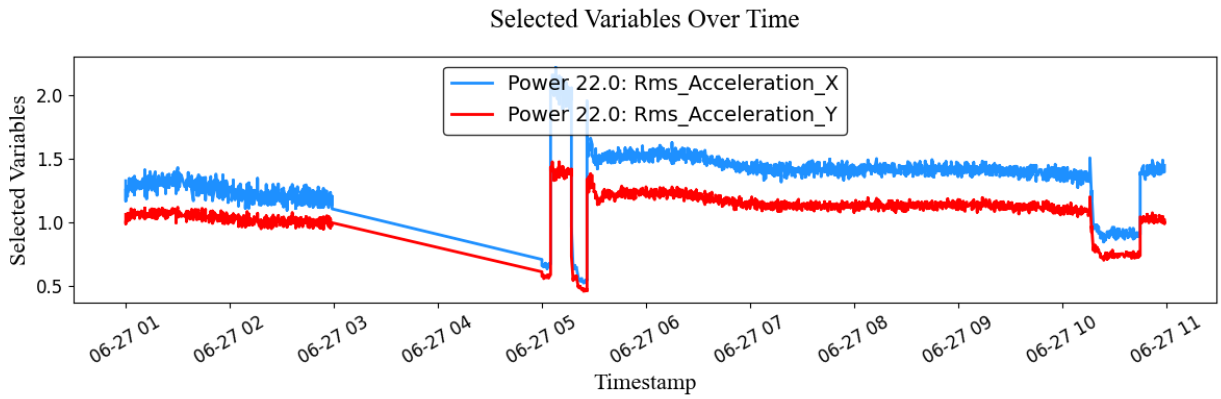
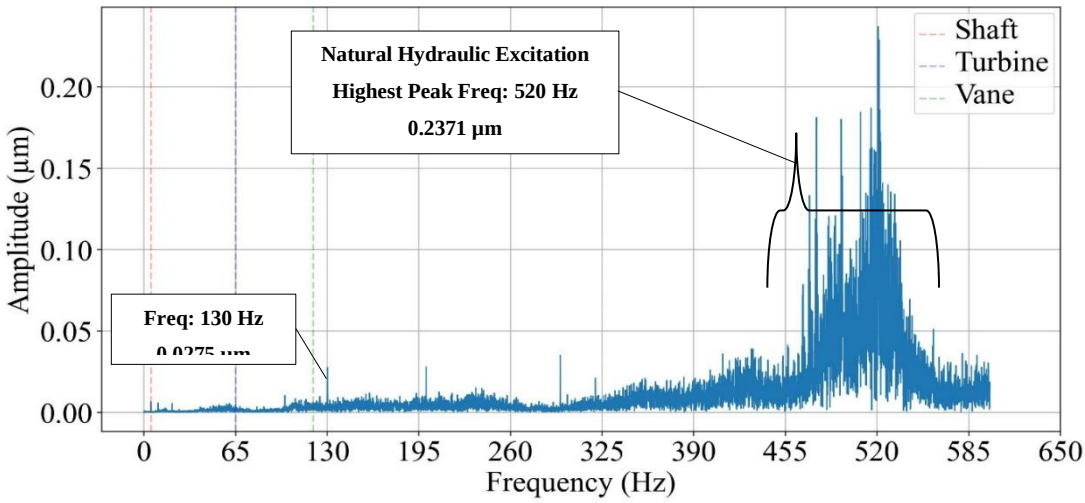


Figure 15: RMS Acceleration of 27th June

- **4.4.1.3 Spectrum Analysis at 24 MW (July 12, 2025)**



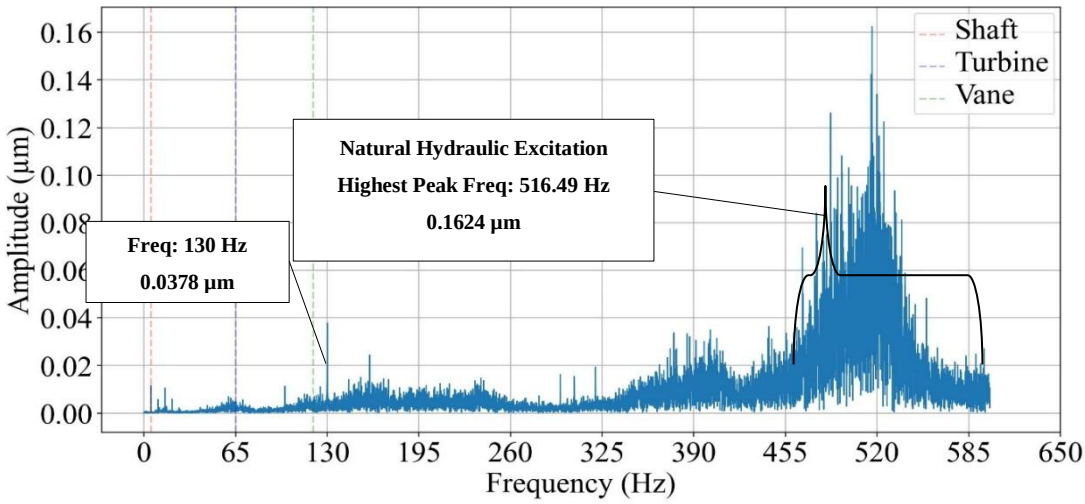


Figure 16: Spectrum Analysis of 12th July at 24 MW Power

As shown in Figure 16, the vibration spectrum at 24 MW load (July 12th, 2025) demonstrates consistent frequency patterns across both the X and Y axis, with shaft and vane passing

frequencies remaining nearly unchanged compared to lower load conditions. The measured vibration levels, ranging from 10–15 µm (peak-to-peak) and an RMS velocity within ISO 20816 recommended limits (0.3–0.5 mm/s), confirm that the unit is operating under a healthy mechanical state. The most significant contributions in the spectra are associated with the turbine rotational frequency and its harmonics, which exhibited higher amplitudes along the X-axis, indicating directional sensitivity of the vibration response. Notably, the increase in amplitude correlates with the higher operating load of 24 MW, which intensifies rotor–stator interactions within the turbine flow passage.

A strong natural hydraulic excitation is evident in the 450–600 Hz frequency band, where pronounced high-frequency activity is linked to flow-induced vibrations. This is consistent with the slightly elevated draft tube pressure head of 1.55 m, compared to the 22 MW case. The increased head enhances the excitation of structural resonances, amplifying the higher frequency components. Despite these effects, shaft and vane frequencies remained stable, while the turbine frequency and its harmonics clearly dominated the overall response.

The analysis highlights that although the system experiences stronger hydraulic excitation at higher loads, the machine continues to operate within acceptable vibration standards. The dominance of turbine harmonics in the X-axis spectrum, coupled with stable shaft and vane frequencies, confirms that the plant is in good operating condition, with no signs of abnormal mechanical or hydraulic instabilities. This behavior underscores the direct influence of increased load and head on excitation mechanisms while reaffirming the robustness of the unit under rated conditions.

- **4.5.1.4 Comparative Vibration Spectrum Analysis**

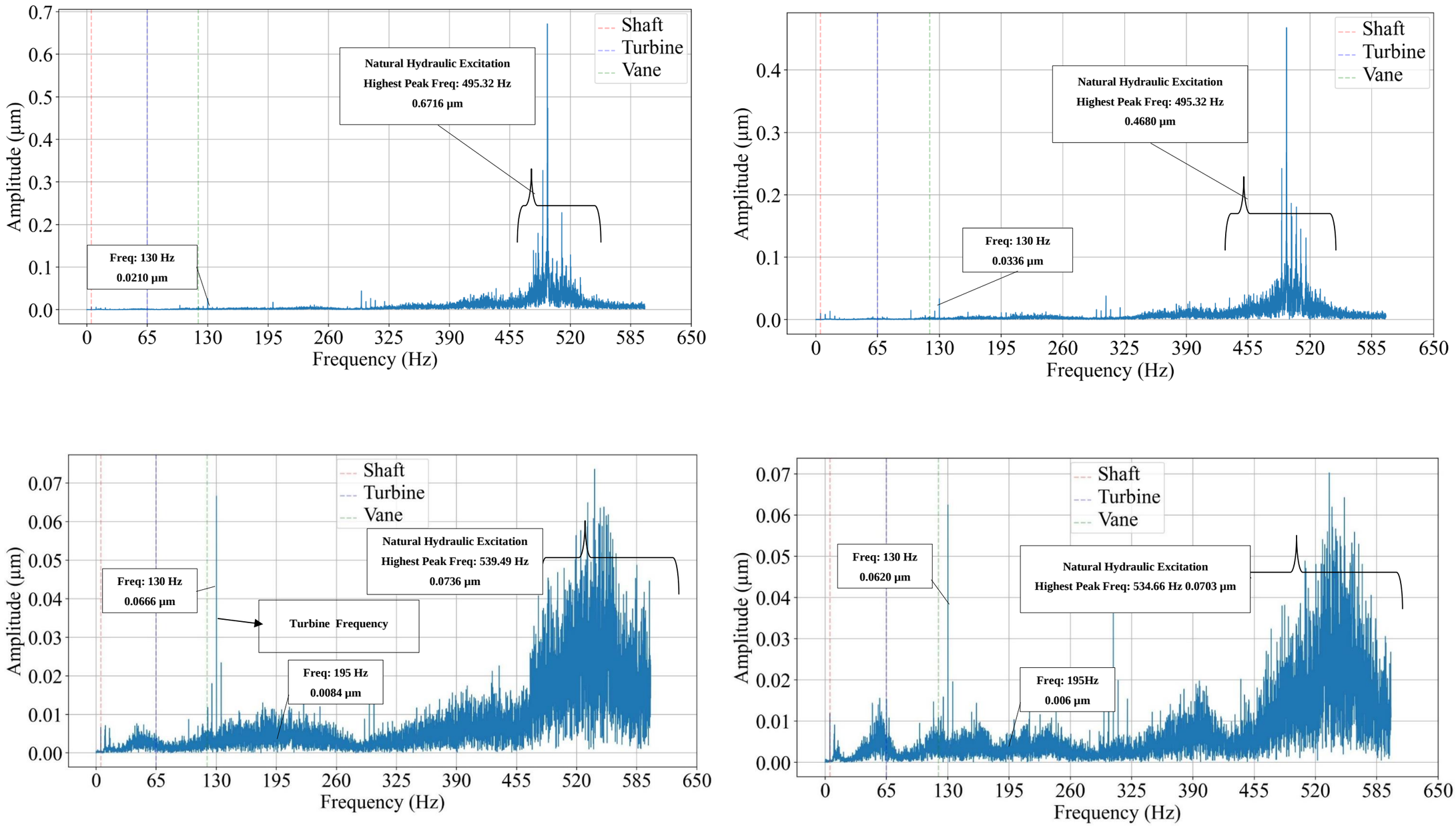


Figure 17: Spectrum Analysis of 12th July and 22nd July at 22 MW Power

The above graph in Figure 17 shows a comparison of spectra taken at different times for the same power level of 22 MW. When examining the spectra from July 12 and July 22, observed changes in the amplitude at the turbine frequency, while the shaft and vane passing frequencies remained nearly unchanged, showing no significant variation.

These measurements were recorded as peak-to-peak displacement in both the X and Y axis, with vibration levels ranging from 10 µm to 15 µm. The RMS velocity, which indicates the strength of vibration, remained within ISO guidelines (0.3 to 0.5 mm/s), indicating that the plant is in good condition.

- **The increase in turbine frequency amplitude**

On the July 12 spectrum, the turbine frequency showed clear peaks. For the X-axis vibration, the amplitude at the turbine frequency, 130 Hz, was approximately 0.0210 µm, while on the Y-axis it reached about 0.0336 µm. By July 22, these amplitudes had increased further, as seen in the spectrum. The X-axis turbine frequency amplitude, 130 Hz, was around 0.0666 µm, and the Y-axis amplitude was about 0.0620 µm. This growth in turbine frequency amplitude and its surrounding harmonics indicates increased dynamic response of the turbine system during this period. The rise is likely due to operating or structural factors that enhanced vibration transmission from the turbine to the shaft and bearings.

- **The natural Hydraulic Excitation**

Notably, vibrations in the range of 450 to 600 Hz were observed, which corresponds to natural hydraulic excitation, also known as flow-induced vibration. This phenomenon represents the transition from random hydraulic excitation to structural resonance at a specific natural frequency. These effects were particularly evident when the plant was running continuously at 22 MW for longer time and sudden change of power to 18 MW for 2 hours and during the transition to 17 MW power this frequency are seen. During these measurements, the draft tube pressure was around 1.5 m under stable conditions. By July 22, it had slightly increased to 1.6 m. This 0.1 m rise effectively increases the plant head by about 1 m, raising the tailrace water level. While this boosts power output, it also introduces stronger high-frequency vibrations called Natural hydraulic excitation.

- **Comparative analysis**

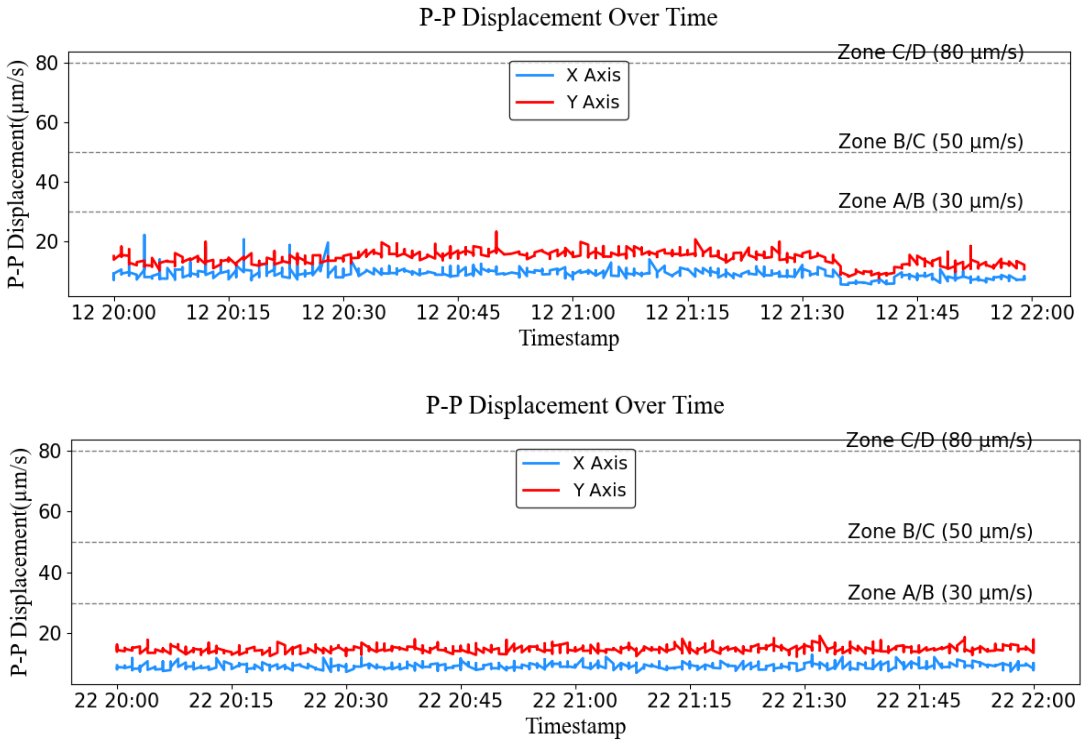


Figure 18: P-P Displacement of 12th July and 22nd July Measurement at 22MW

As shown in Figure 18 July 12 measurement illustrates that the X-axis displacement fluctuates primarily within the range of 8–12 µm, whereas the Y-axis displacement shows slightly higher values, averaging between 10–15 µm with occasional peaks surpassing 18 µm. Toward the later part of the observation period, a mild stabilization trend is visible, with both X and Y displacements shifting toward their lower bands. This suggests that the machine was operating in a steady state after experiencing minor fluctuations. Importantly, the recorded displacements remain significantly below the ISO 20816 Zone A/B guideline of 30 µm, confirming that the equipment was running under favorable vibration conditions. The observed variations may be linked to transient operational influences or minor system adjustments, which, although not alarming, merit attention for trend analysis.

In comparison, the July 22 measurement exhibits a slightly more consistent vibration pattern. The X-axis displacement predominantly lies between 7–10 µm, while the Y-axis remains in the 11–14 µm band with occasional excursions beyond 15 µm. Unlike the July 12 dataset, the fluctuations are less pronounced, and the displacement values demonstrate a relatively stable trajectory throughout the observation window. This consistency reflects smoother machine dynamics, possibly indicating reduced operational disturbances or improved system balance

on this date. Again, all measured values are comfortably within the ISO 20816 Zone A/B range, confirming safe and acceptable vibration performance.

Taken together, the results from both July 12 and July 22 indicate that the machine was consistently operating well below the critical ISO 20816 threshold, demonstrating a good vibration health status at 22 MW load. The slight transient variations observed across the two datasets are not indicative of fault conditions but rather highlight the presence of minor operational or system adjustments. Such findings reinforce the reliability of the machine's performance while underscoring the value of continued monitoring to detect early signs of evolving mechanical changes.

- **Comparison of July 12th and July 22nd RMS Velocity Behavior with ISO 20816**

As shown in Figure 19, the July 12 RMS velocity trend indicates that both X and Y axes remain

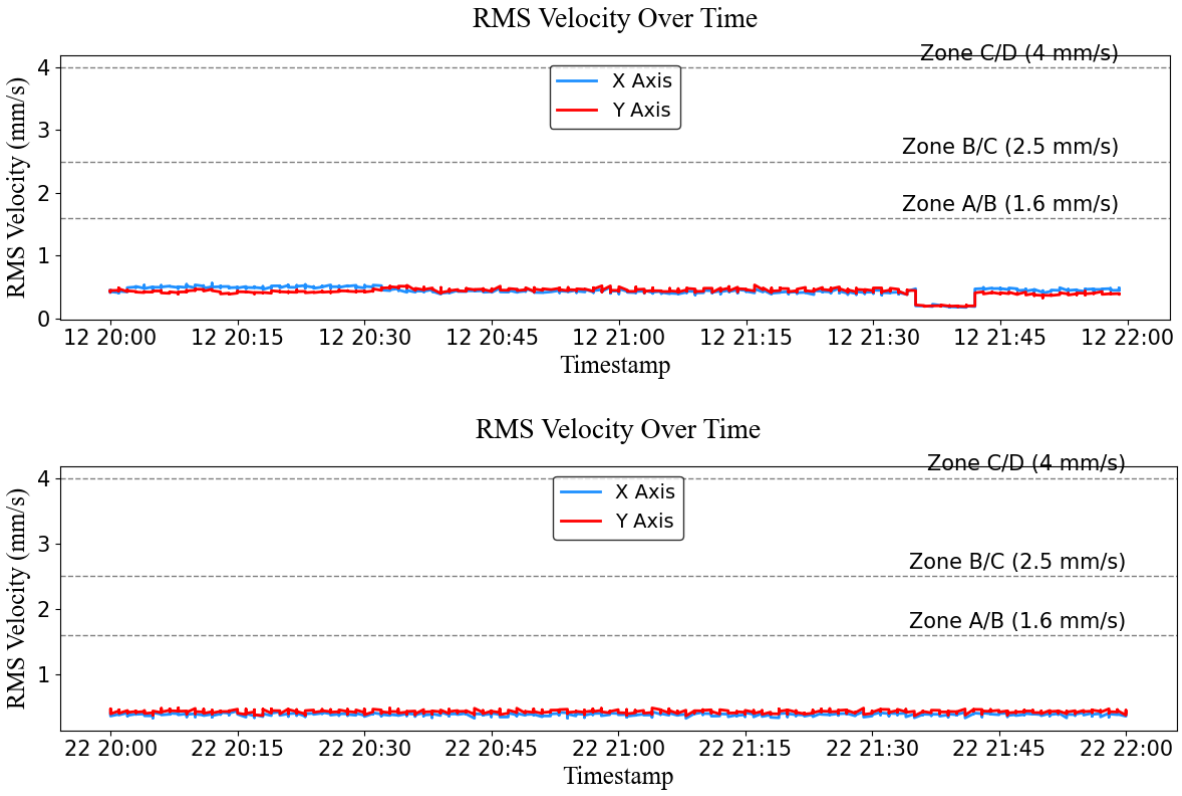


Figure 19: RMS Velocity of 12th July and 22nd July Measurement at 22MW

stable within the lower range of 0.4–0.6 mm/s, with only minor fluctuations observed throughout the measurement period. A brief reduction is noted around 21:30, where the Y-axis drops close to 0.2 mm/s before recovering to its earlier level near 21:45. Despite this temporary reduction, the velocities across both axes consistently remain well below the ISO 20816 Zone A/B threshold of 1.6 mm/s, confirming that the machine was operating under very safe vibration conditions. The short-term drop and subsequent recovery suggest a transient

operational adjustment or minor disturbance, though not indicative of any mechanical abnormality.

In comparison, the July 22 dataset presents an even more consistent vibration profile. Both X and Y axis velocities remain relatively steady between 0.3–0.5 mm/s across the entire observation window, with minimal deviations. Unlike the July 12 case, no distinct transient event is evident, and the machine demonstrates smoother operational stability. The consistently low velocity levels, again well within the ISO 20816 Zone A region, reinforce the conclusion that the machine was functioning in a healthy and controlled state during this measurement.

Overall, the comparison between July 12 and July 22 highlights that the machine maintained very low RMS velocity values on both occasions, reflecting excellent vibration health and operational reliability. While July 12 exhibited a short-lived disturbance, both datasets confirm operation far below critical ISO 20816 thresholds, signifying a robust condition with no signs of developing faults. Continued monitoring is still recommended to capture early signs of any emerging changes, but the present analysis confirms the machine's strong vibration performance at 22 MW load.

- **4.5.1.5 Comparative study of the spectrum of 26th July and 8th August on 20MW Power**

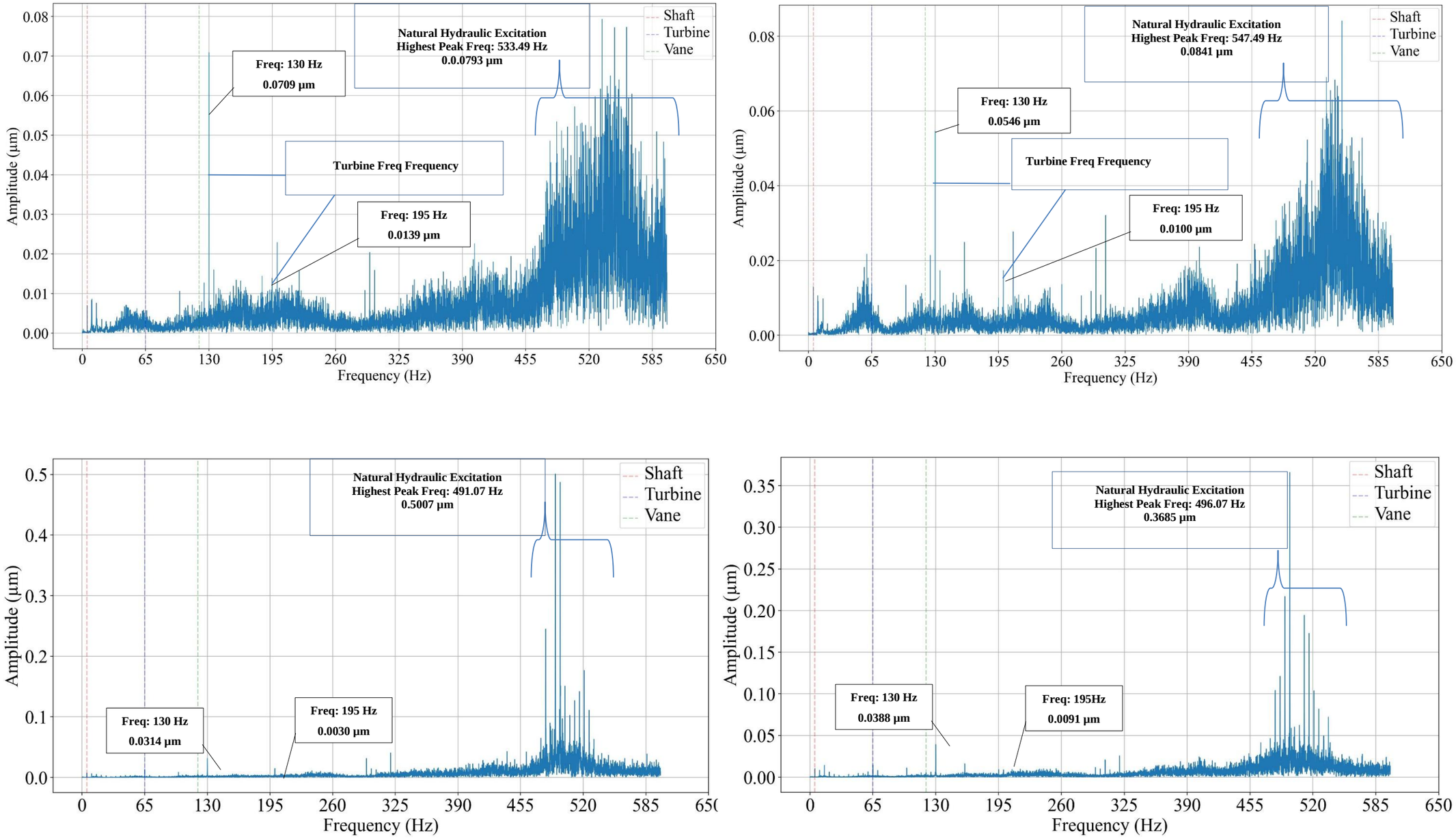


Figure 20: Spectrum Analysis of 26th July and 8th August at 20 MW Power

The above Figure 20 plot shows a comparison of spectra taken at different times for the same power level of 20 MW. When examining the spectra from July 26 and August 8, observed changes in the amplitude at the turbine frequency, while the shaft and vane passing frequencies remained nearly unchanged, showing no significant variation.

These measurements were recorded as peak-to-peak displacement in both the X and Y axis, with vibration levels ranging from 10 µm to 15 µm. The RMS velocity, which indicates the strength of vibration, remained within ISO guidelines (0.3 to 0.5 mm/s), indicating that the plant is in good condition.

- **The increase in turbine frequency amplitude**

On the July 26 spectrum, the turbine frequency and its harmonics showed increased amplitudes. For the X-axis vibration, the first harmonic of the turbine frequency was 0.0709 µm, and the second harmonic was 0.0139 µm. For the Y-axis vibration, the first harmonic was 0.0546 µm, and the second harmonic was 0.0100 µm, which can also be seen in Figure 10. By August 8, these peaks had decreased, indicating stabilization. This is due to the higher turbine frequency observed in July

- **The natural Hydraulic Excitation**

Notably, vibrations in the range of 450 to 600 Hz were observed, which corresponds to natural hydraulic excitation, also known as flow-induced vibration. This phenomenon represents the transition from random hydraulic excitation to structural resonance at a specific natural frequency. These effects were particularly evident when the plant was running continuously at 22 MW for a longer time and a sudden change of power to 18 MW for 2 hours, and during the transition to 18 MW power, these frequencies are seen. During these measurements, the draft tube pressure was around 1.5 m under stable conditions. By August, it had slightly increased to 1.6 m. This 0.1 m rise effectively increases the plant head by about 1 m, raising the tailrace water level. While this boosts power output, it also introduces stronger high-frequency vibrations called Natural hydraulic excitation.

- **Comparison of July 26 and August 8 Displacement Behavior with ISO 20816**

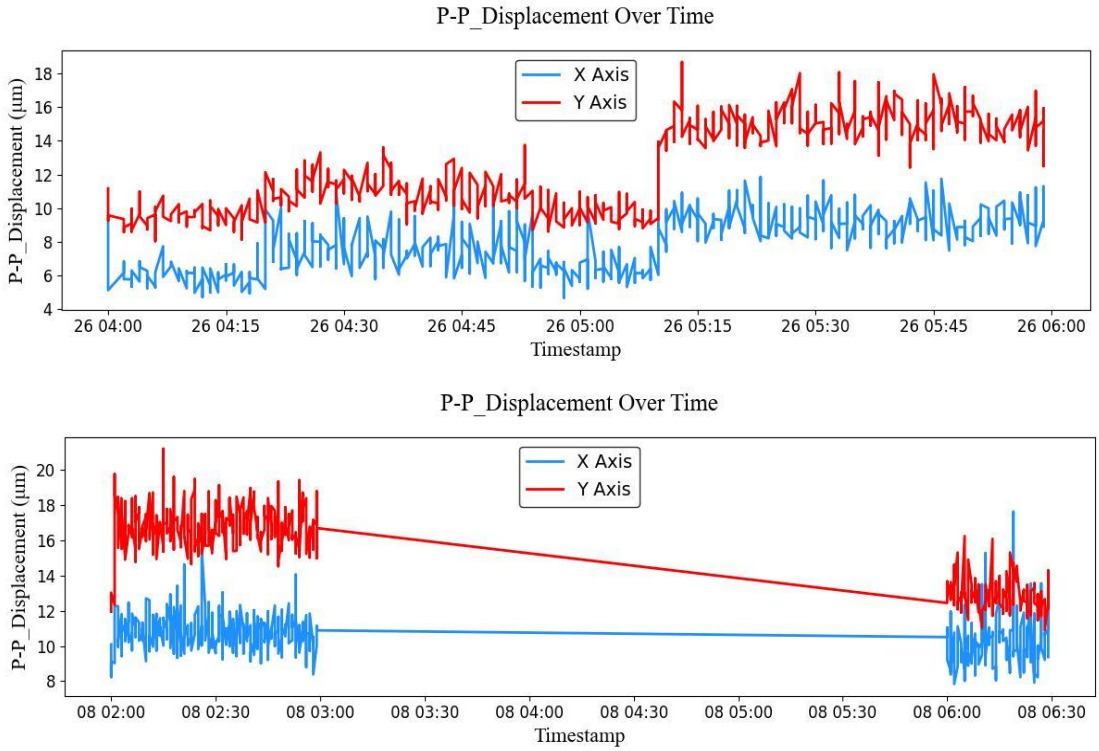


Figure 21: P-P Displacement of 26th July and 8th August Measurement

The July 26 plot shows that the X-axis displacement fluctuates between 5–10 µm and the Y-axis displacement ranges from 9–12 µm in the early period, followed by a step increase after 05:05 AM. Post-event, the X-axis stabilizes closer to 8–10 µm, while the Y-axis rises more significantly to 13–16 µm. These values remain well below the ISO 20816 threshold of 30 µm for Zone A/B, meaning the machine was still operating within a good vibration condition despite the noticeable increase. However, the sudden step change in amplitude suggests that some disturbance or change in machine operation occurred, which, although not critical according to ISO standards, is an early indicator of possible evolving issues. This issue can be highlighted through the frequency graph analysis.

In contrast, the August 8 plot initially records significantly higher displacements, fluctuating between 12–18 µm before 08:03:00 AM. Although still within Zone A/B (<30 µm), the vibrations were relatively elevated compared to July 26. After 08:03:00 AM, both X and Y axis show a sharp drop, stabilizing around 2–4 µm. This reduction places the system in an excellent vibration condition well inside Zone A, reflecting either successful system stabilization, mechanical realignment, or control intervention. The low stable values after the drop suggest

the system entered a much healthier and more reliable operating state compared to its pre-drop condition.

However, the analysis of vibration displacement and frequency amplitude from both July 26 and August 8 shows that there is no significant increase in either displacement levels or frequency amplitude. All recorded values remain well within the ISO 20816 guideline limits, falling under Zone A, which represents a good vibration condition. This indicates that the machine is operating safely, with no critical abnormalities, and the components are in good condition.

CHAPTER 5 CONCLUSION

5.1 Determining the Vibrational Characteristics of the Francis Turbine at the Lower Marshyangdi Hydropower Plant at Different Operational Conditions

To determine the vibrational characteristics of the Francis turbine at the Lower Marshyangdi Hydropower Plant under different operational conditions, the team conducted a comprehensive case study focusing on part load (16–18 MW) and full load (21–24 MW) scenarios. The study began by reviewing existing literature on vibration sources in Francis turbines, such as hydraulic forces from turbulence and pressure fluctuations, mechanical issues like shaft imbalance and misalignment, and electromechanical effects from generator imbalances. At the plant, uniaxial piezoelectric accelerometers were installed on the turbine guide bearing casing to capture radial vibrations in the X and Y axes. Data was collected in real-time using a National Instruments (NI) Data Acquisition (DAQ) system and vibration module, with signals sampled at 12,800 Hz and processed through LabVIEW for initial monitoring. The raw acceleration data was then filtered using Butterworth high-pass and low-pass filters to remove noise and resonance, applying the Nyquist theorem to eliminate aliasing. The signals were converted from the time domain to the frequency domain via Fast Fourier Transform (FFT) in Python, identifying key frequencies: shaft (~5 Hz), turbine (~65 Hz), and vane passing (~120 Hz). Velocity and displacement were derived through concatenated integration of acceleration data. These signatures were analyzed against ISO 20816 guidelines, evaluating RMS velocity (0.3–0.5 mm/s) and peak-to-peak displacement (10–15 µm), which remained within safe Zone A/B limits. Comparative analyses across dates (e.g., July 12 vs. July 22 at 22 MW) revealed stable frequencies with minor amplitude increases in turbine harmonics due to draft tube pressure variations (1.5–1.6 m), highlighting flow-induced excitations in the 450–600 Hz range. This work confirmed the turbine's healthy post-overhaul state, with vibrations intensifying slightly at higher loads due to enhanced rotor-stator interactions, enabling predictive insights for maintenance in sediment-prone environments.

5.2 Studying the Vibrational Characteristics of the Francis Turbine

In studying the vibrational characteristics of the Francis turbine, the study delved into the fundamental principles of mechanical vibrations, modeling them using concepts like mass-spring-damper systems governed by differential equations involving mass (M), damping coefficient (C), stiffness (K), displacement (X), and external forces (F(t)). Classifications such

as free versus forced vibrations, damped versus undamped, and natural frequency with resonance risks were explored. Specific to Francis turbines, the study examined sources like hydraulic instabilities (e.g., vortex shedding at part load), mechanical faults (e.g., erosion from Himalayan sediment reducing blade thickness by up to 5 mm over the years), and electromechanical imbalances. Literature reviews covered techniques like FFT for frequency-domain analysis, alongside alternatives such as Hilbert–Huang Transform (HHT) and Wavelet Transform (WT) for non-stationary signals. Operational zones were analyzed: Best Efficiency Point (BEP) with minimal vibrations, part load with increased turbulence, and full load with transient excitations. Broadband vibrations from sediment-laden flows were noted, and ISO 20816-5 standards were applied to define severity zones (A–D) based on RMS velocity and peak-to-peak displacement thresholds. Through this, harmonics were identified as indicators of faults, emphasizing condition-based monitoring for predictive maintenance to enhance reliability and efficiency in hydropower plants like Lower Marshyangdi.

5.3 Determining the Instrumentation, Position, and Installation of the Vibration Sensor

To determine the instrumentation, position, and installation of the vibration sensor, the team first evaluated various options based on literature and site constraints at the Lower Marshyangdi Hydropower Plant. A uniaxial piezoelectric accelerometer with an 8 kHz frequency range and 101 mV/g sensitivity was selected for its suitability in detecting high-frequency vibrations in tight spaces. Alternatives like proximity transducers and seismic sensors were considered but deemed less ideal due to space limitations at the turbine guide bearing casing. The position was chosen on the turbine guide bearing for optimal radial vibration capture in X and Y axes at 90° angles, informed by studies recommending placements near bearings, draft tubes, and guide vanes for comprehensive monitoring. Installation involved drilling a hole matching the sensor's diameter for a tight interference fit, ensuring secure mounting to avoid signal distortion. The sensor was integrated with an NI C-series 4-channel vibration module for signal conditioning (amplification and filtering) and a DAQ system for digitization. LabVIEW was used to create a graphical interface for real-time control and data visualization, allowing monitoring of electrical signals converted to acceleration (in g). This setup was tested on-site with administrative support from plant staff, confirming accurate data transmission to a server at Kathmandu University for further processing.

5.4 Storing and Analyzing Vibration at Different Operating Conditions

For storing and analyzing vibration data at different operating conditions, the team established a robust system starting with real-time data collection via the installed accelerometer, vibration module, and DAQ, logging signals in formats like CSV or TDMS through LabVIEW. Data was securely stored on a server at Kathmandu University, organized by date, load (part: 16–18 MW; full: 21–24 MW), and axis (X/Y) for easy retrieval. Analysis began with preprocessing in Python: applying Butterworth bandpass filters to isolate relevant frequencies (10–1000 Hz for radial vibrations), removing noise, and preventing aliasing per the Nyquist criterion. FFT was performed to generate frequency spectra, plotting amplitude versus frequency to identify peaks at shaft (5 Hz), turbine (65 Hz), and vane passing (120 Hz) frequencies. Acceleration was integrated to compute velocity (for severity) and displacement (for movement extent) using first- and second-order equations, with high-pass filtering to correct baseline drift. Trends like RMS velocity and peak-to-peak displacement were evaluated against ISO 20816 thresholds, conducting comparative studies (e.g., June 25 vs. July 4 at 21 MW; June 26 vs. 27 at 22 MW) revealing hydraulic resonance from draft tube pressure changes. Broadband high-frequency content (450–600 Hz) was linked to flow-induced excitations, with visualizations in time and frequency domains supporting fault diagnosis. This process enabled condition-based predictive maintenance recommendations, ensuring turbine reliability across operational variations.

CHAPTER 8 REFERENCES

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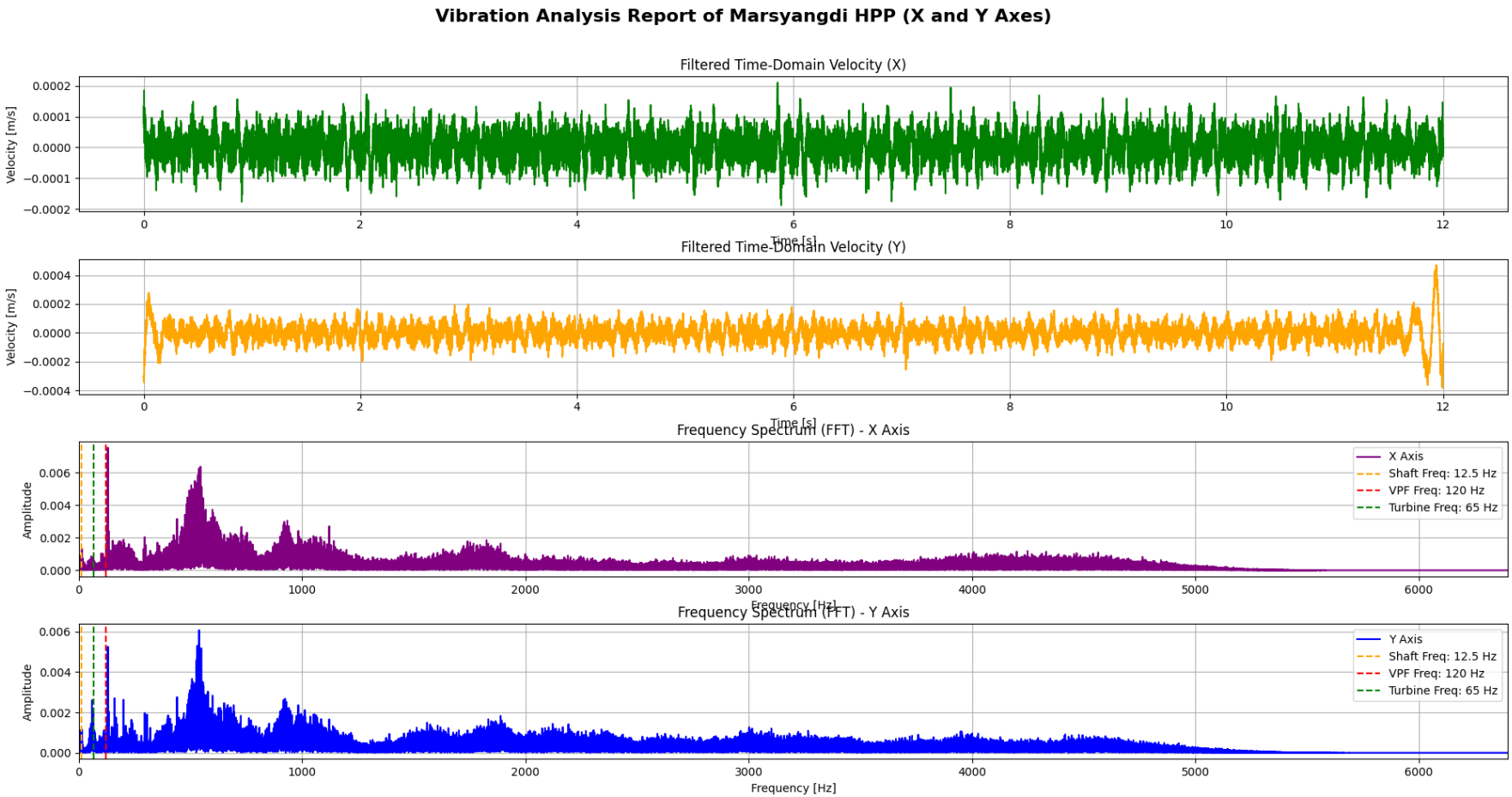
ANNEX


Vibration Analysis Report of Marsyangdi HPP (X and Y Axes)
Filtered Time-Domain Velocity (X)
Velocity [m/s]
Time [s]
Filtered Time-Domain Velocity (Y)
Velocity [m/s]
Time [s]
Frequency Spectrum (FFT) - X Axis
Amplitude
Frequency [Hz]
X Axis
Shaft Freq: 12.5 Hz
VPF Freq: 120 Hz
Turbine Freq: 65 Hz
Frequency Spectrum (FFT) - Y Axis
Amplitude
Frequency [Hz]
Y Axis
Shaft Freq: 12.5 Hz
VPF Freq: 120 Hz
Turbine Freq: 65 Hz

Vibration Measurement at Marsyangdi Hydropower Station -- Unit 2

Sat, 09 Aug 2025 23:31:36

Pressure

Inlet: 9.5 bar

Draft Tube: 1.68506 bar

X Axis Vibration

RMS Acceleration: 1.15808 m/s²

RMS Velocity: 0.44568 mm/s — Zone A/B (≤ 1.6)

P-P Displacement: 12.9128 µm — Zone A/B (≤ 30)

Y Axis Vibration

RMS Acceleration: 1.12665 m/s²

RMS Velocity: 0.512427 mm/s — Zone A/B (≤ 1.6)

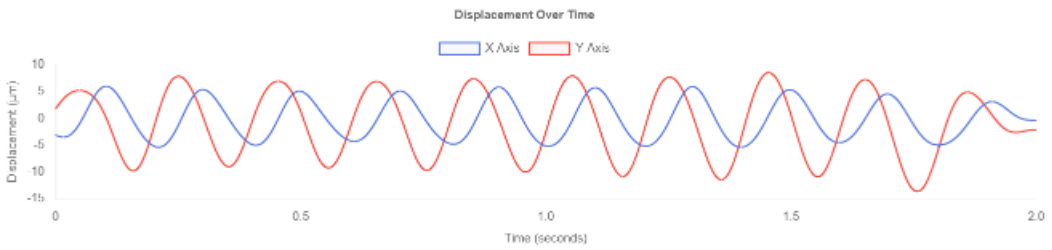
P-P Displacement: 19.3702 µm — Zone A/B (≤ 30)

Legend

Zone	RMS Velocity (mm/s)	P-P Displacement (µm)
Zone A/B	≤ 1.6	≤ 30
Zone B/C	≤ 2.5	≤ 50
Zone C/D	≤ 4.0	≤ 80
Above D	> 4.0	> 80

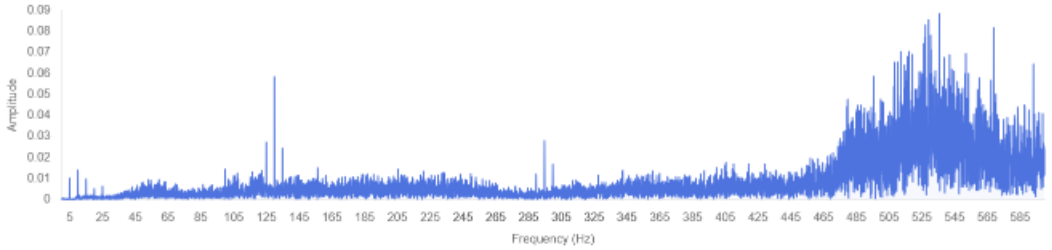
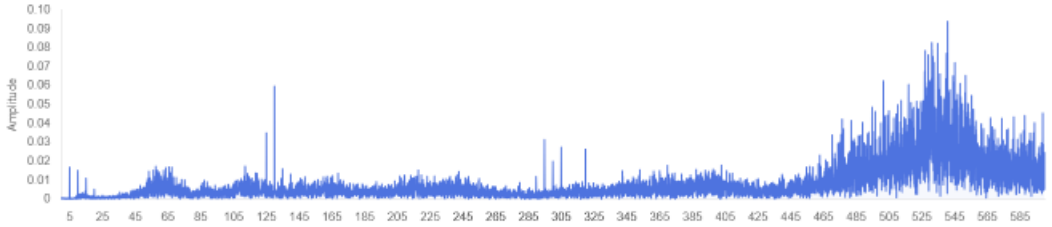
Unit Operation Status

Shaft Displacement Graph



Frequency Spectrum (Sat, 09 Aug 2025 23:31:49)

X-Direction



Y-Direction